

Introduction

MCP is a standard that allows agents to securely and flexibly interact with external tools and data sources through MCP servers. MCP enables agents to discover, connect to, and start external tools that are provided by MCP servers. These servers function as intermediaries, presenting various capabilities (such as searching the web, querying a database, sending email, or fetching web content) in a standardized manner. This process does not require creating custom APIs or manually setting up connections, making it a simpler and more efficient method for agents to use external tools.

With MCP, agents can discover and use these external tools during run time, enhancing their capabilities beyond their inherent features. All interactions with external tools are managed through a single protocol, which simplifies connections and improves efficiency. Moreover, MCP facilitates collaboration and tool sharing among agents from different vendors or platforms.

During this lab you will learn how to add a Tavily web search tool to an agent, imported using an MCP server provided by Tavily.

Tavily is a search engine and API specifically designed for AI agents and Retrieval-Augmented Generation (RAG) workflows. It provides real-time, accurate, and factual search results tailored for use with large language models (LLMs). Unlike traditional search engines, Tavily focuses on providing concise, ready-to-use information optimized for LLM context, streamlining the process of integrating dynamic web information into AI-driven solutions.

Warnings

When running MCP servers without proper sandboxing, they can execute arbitrary code on your system with the same permissions as the host process. This creates significant security risks.

Security Risks:

- System Access: Full access to files, network, and system resources.
- Code Execution: Can run any command on your machine.
- Prompt Injection: Malicious prompts could trigger unintended server actions.
- Data Exposure: Sensitive data may be accessed or leaked.

Best Practices:

- Ensure that the MCP server is legitimate and safe for your agents to use.
- Run servers in VMs or isolated environments.
- Review code before installation.
- Limit permissions to minimum required.
- Monitor server activity.

Note: Watsonx Orchestrate runs local MCP servers in an isolated VM.

About this lab

Lab overview

This lab will teach you how you can import external tools from Model Context Protocol (MCP) servers and add them to your agents to enhance the agent's ability to accomplish tasks.

This lab is composed of two independent parts:

- Part 1: Creating MCP tools using low-code Agent Builder
- Part 2: Creating MCP tools using the ADK

What you will learn

Once you have completed this lab you will have developed skills enabling you to complete the following tasks:

- Installing servers requiring environment variables.
- Creating and configuring a connection.
- Import a tool from an MCP server.
- Create a MCP tool withing the ADK.

How to get support

If you encounter an issue with this lab, you can request assistance through the following channels:

- Review the [Troubleshooting](#) appendix in this guide first.
- IBMers can use the **#ba-techlcd-support** Slack channel.
- If you are an IBM Business Partner and require assistance, please open a support case at IBM Technology Zone Help.

Prerequisites

Make sure the following prerequisites are satisfied before proceeding with this lab:

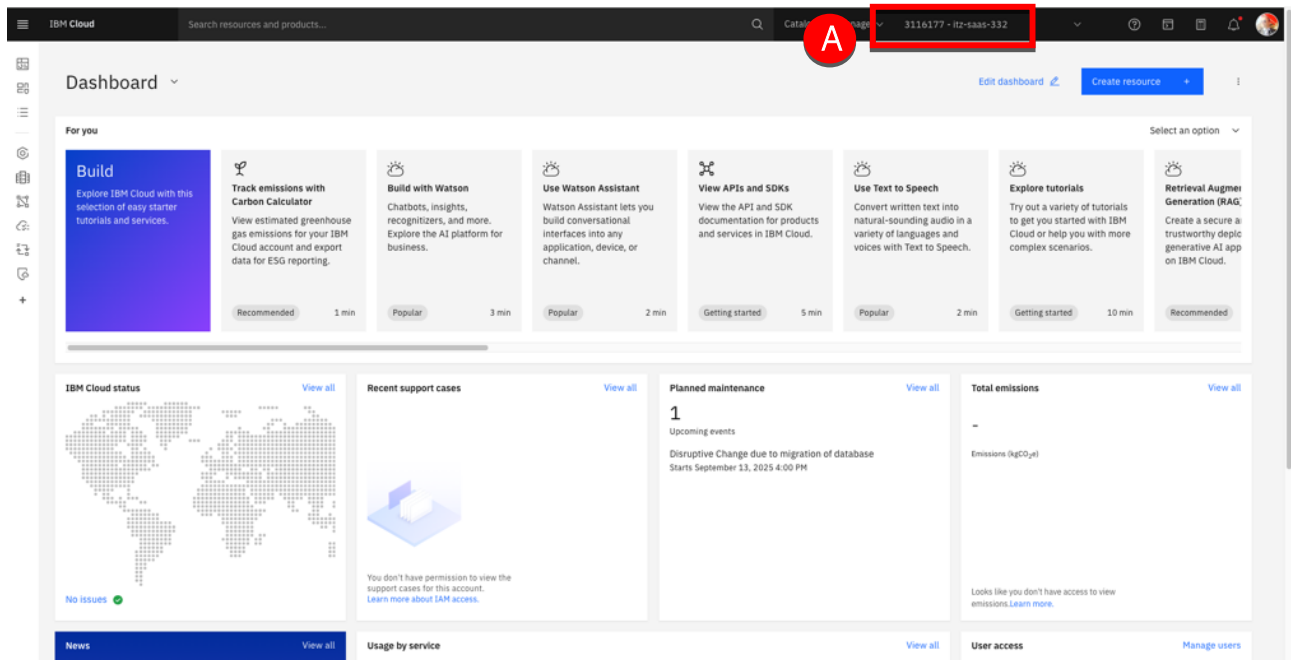
- You have performed the Preparation Lab guide including the environment validation step
- You have your ADK environment up and running (Orchestrate server started after validating the environment in the preparation guide OR after having performed another ADK related lab)

Note: If for some reason your server is not started correctly, reset your server following [the Reset your environment](#) appendix.

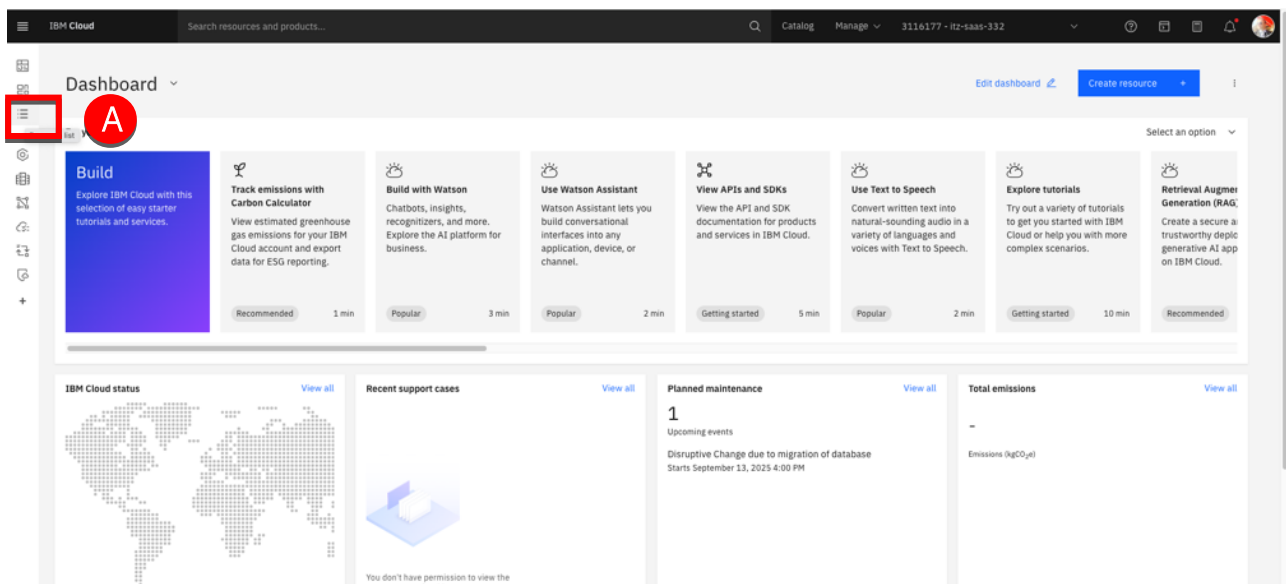
PART 1: Import MCP tools using watsonx Orchestrate

In the first part of this lab, you will learn how to work with MCP tools using watsonx Orchestrate running in IBM Cloud. Refer to your lab credentials in the preparation guide and login to IBM Cloud, ensure you are using the correct account, see item 6 in the lab credentials.

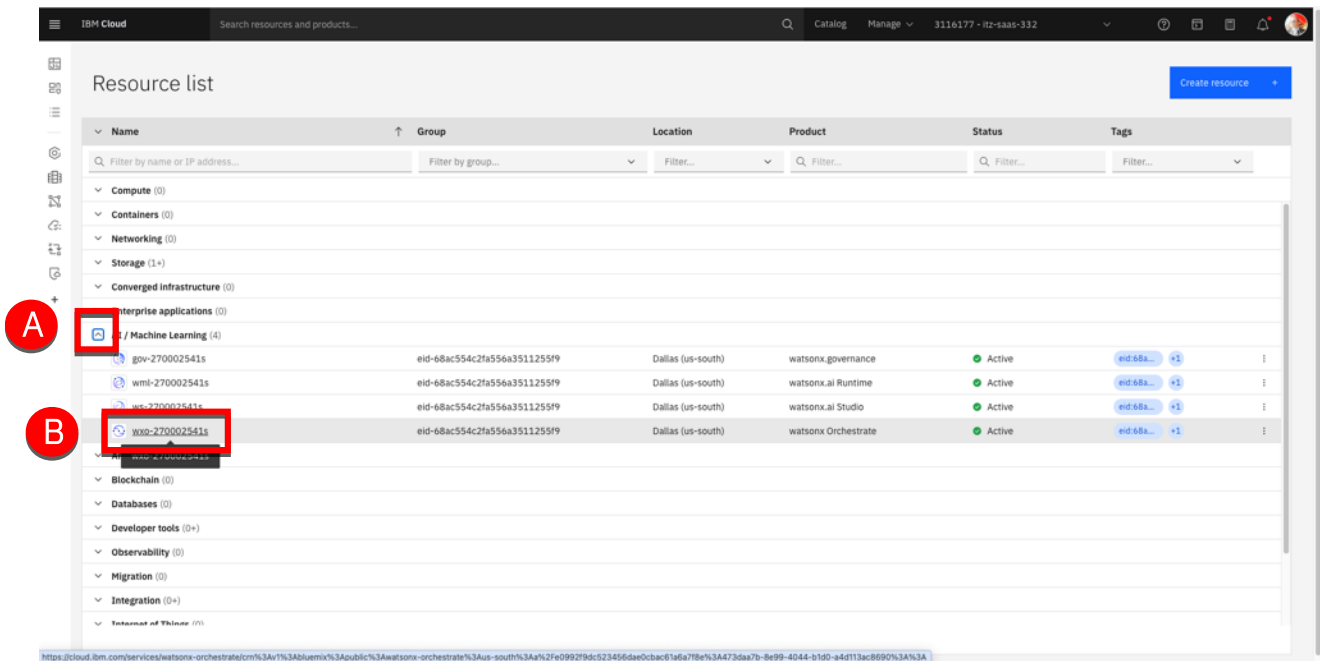
1. Make sure you are on the right cloud account as noted in item 6 in your lab credential notes. This is the IBM Cloud account named in the invitation email you received.



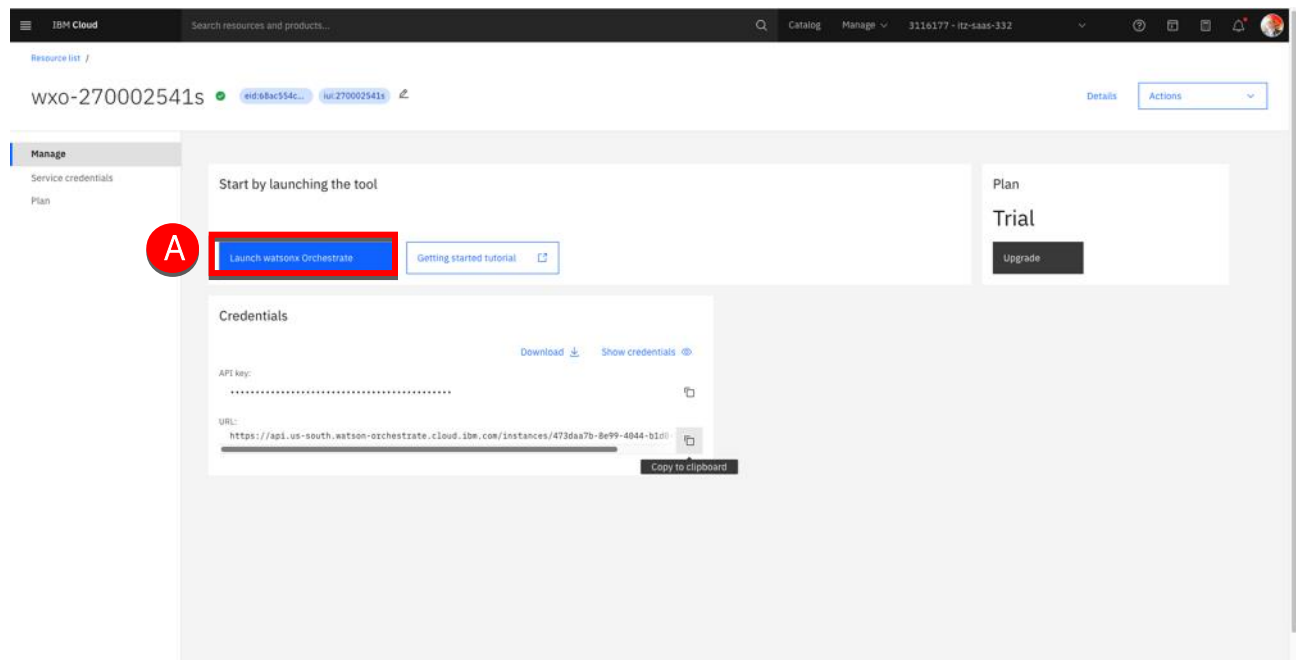
2. Click the Resource list (A).



3. Expand **AI/Machine learning** (A) then click **watsonx Orchestrate** (B).



4. Click **Launch watsonx Orchestrate** (B).



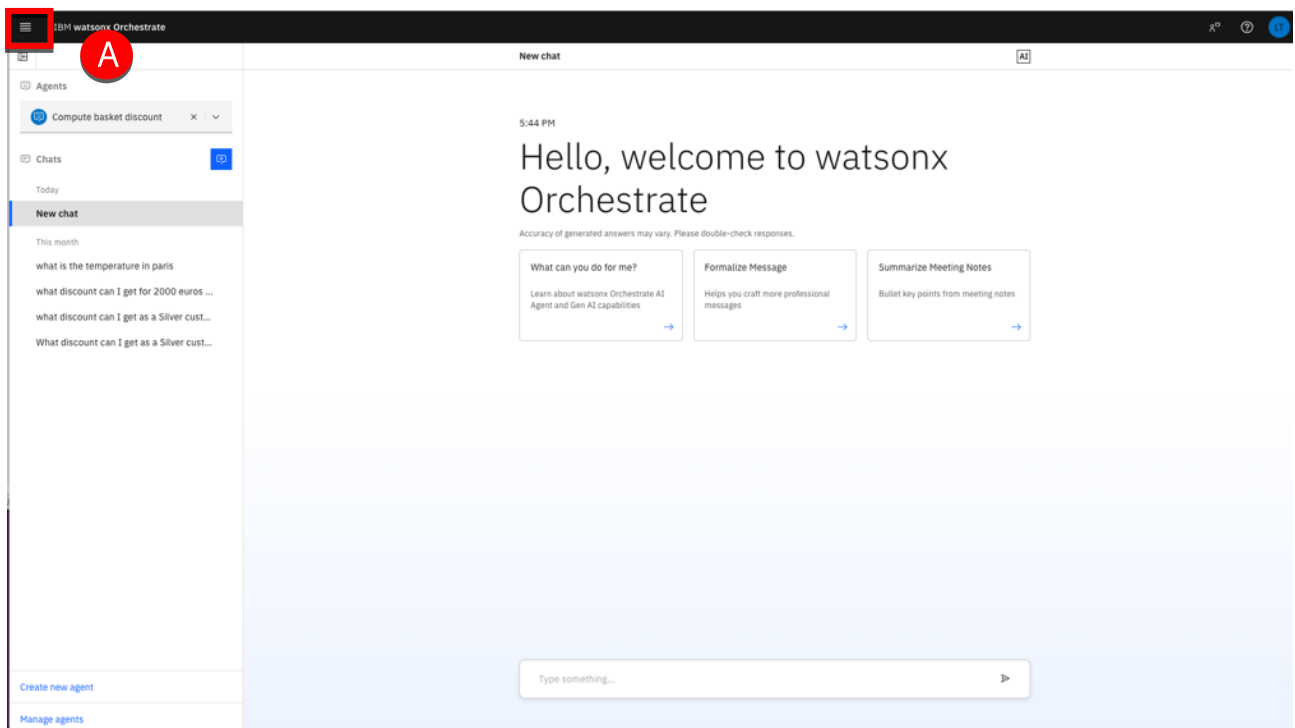
Create a Tavily connection

To install MCP servers that require environment variables, you must add the required variables into an AppID/Connection using key-value pairs. MCP servers fetch the environment variables when they start, so it is important to set the variables before you import or start an MCP server.

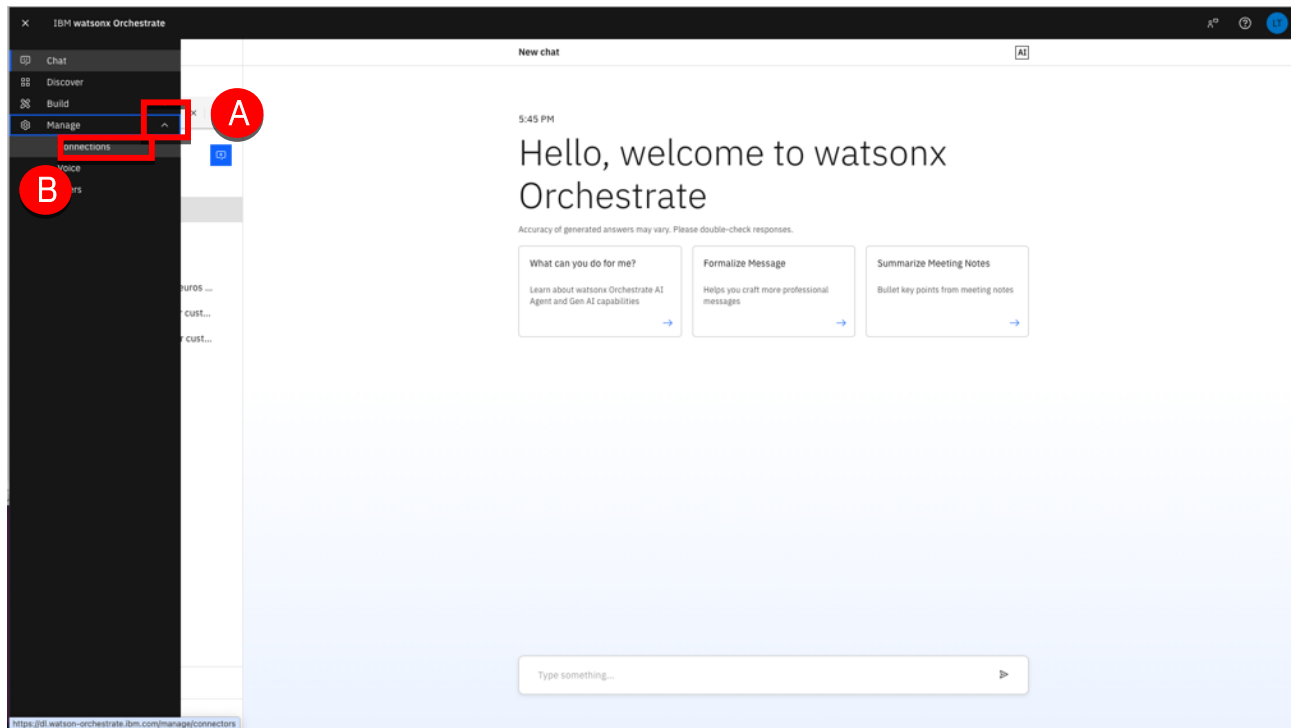
To connect to the Tavily server with your credentials, you must create a new connection in watsonx orchestrate. This is the most secured way to import tools not displaying your API keys and credentials publicly.

Return to your watsonx Orchestrate top menu.

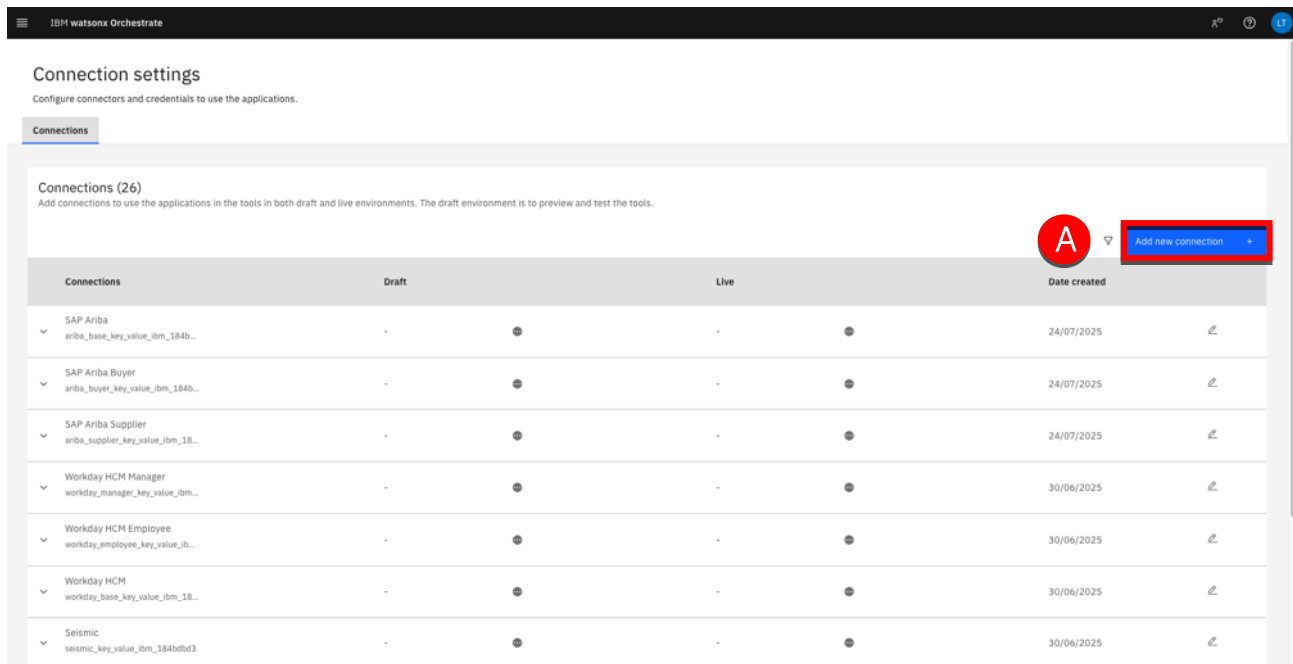
1. Click the Main menu icon (A).



- Expand the **Manage** menu (A) then select **Connections** (B).



- Click **Add new connection** (A).



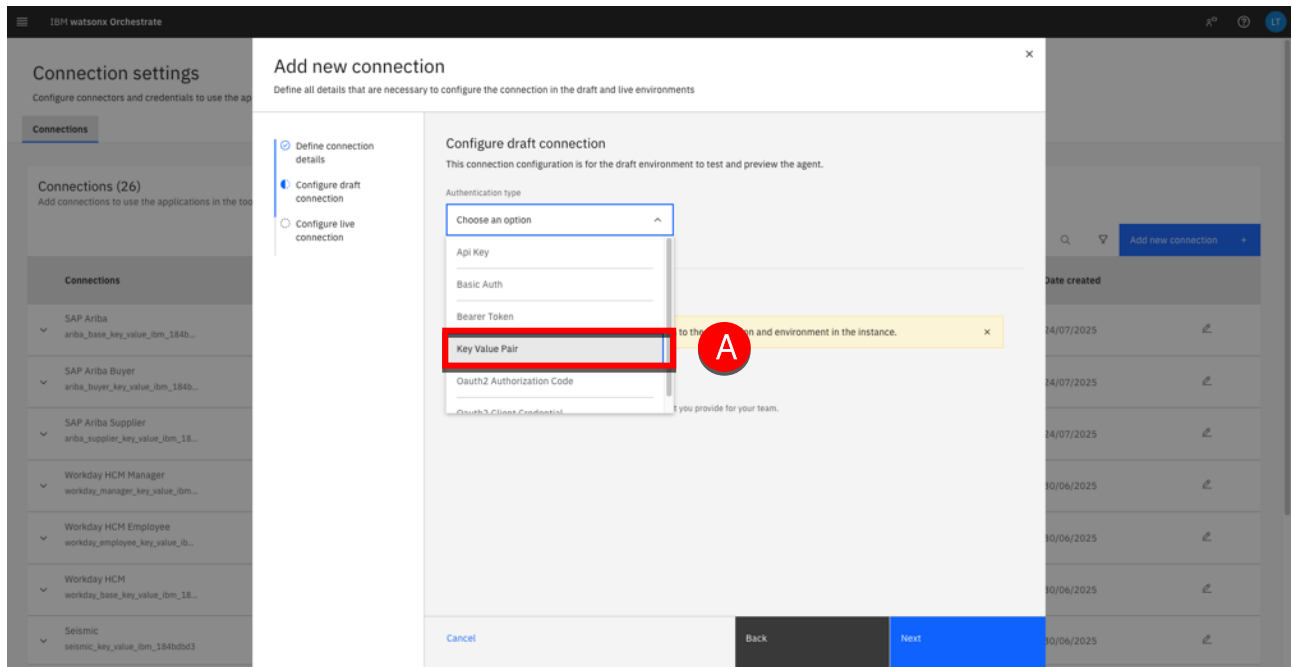
4. Enter Tavily_MCP in the **Connection ID*** field (A), click **Save and continue** (B).

The screenshot shows the 'Add new connection' dialog in IBM watsonx Orchestrate. The 'Define connection details' step is selected. The 'Connection ID*' field is highlighted with a red box and labeled 'A', containing the text 'TAVILY_MCP'. The 'Display name' field is empty. The 'Save and continue' button is highlighted with a red box and labeled 'B'.

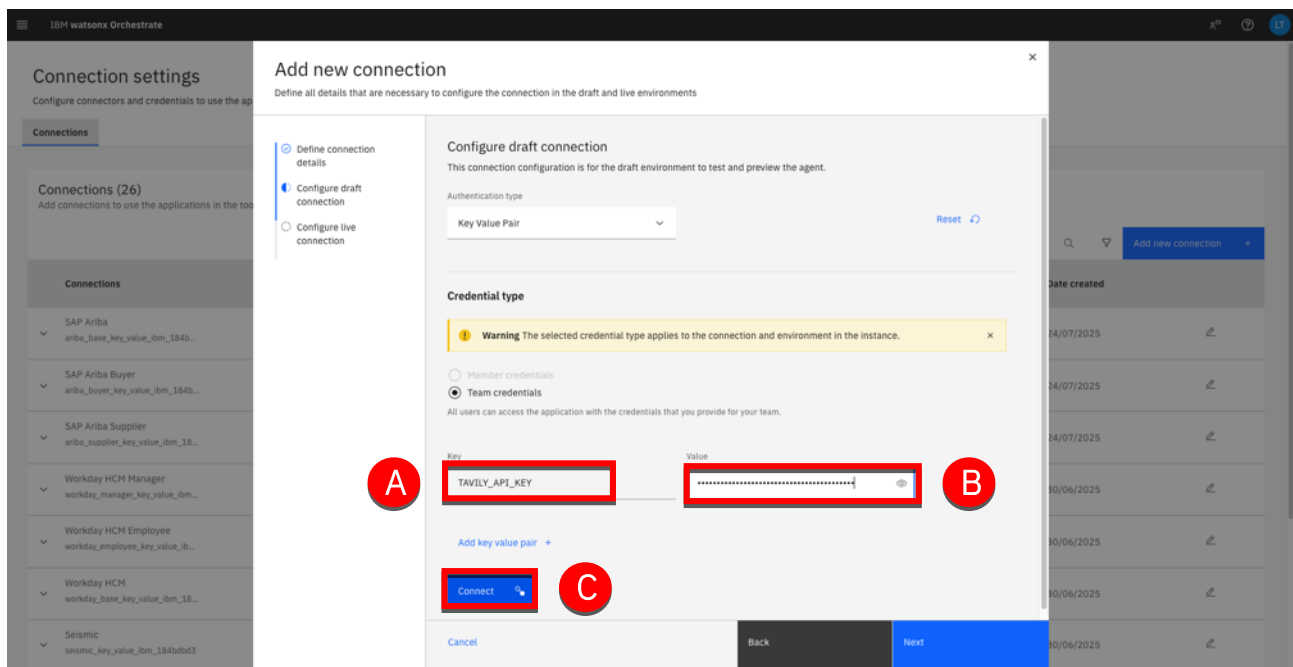
5. Expand the Choose an option in the **Authentication type** menu (A).

The screenshot shows the 'Add new connection' dialog in IBM watsonx Orchestrate. The 'Configure draft connection' step is selected. The 'Authentication type' dropdown menu is highlighted with a red box and labeled 'A', showing the text 'Choose an option'. The 'Credential type' section shows 'Member credentials' selected.

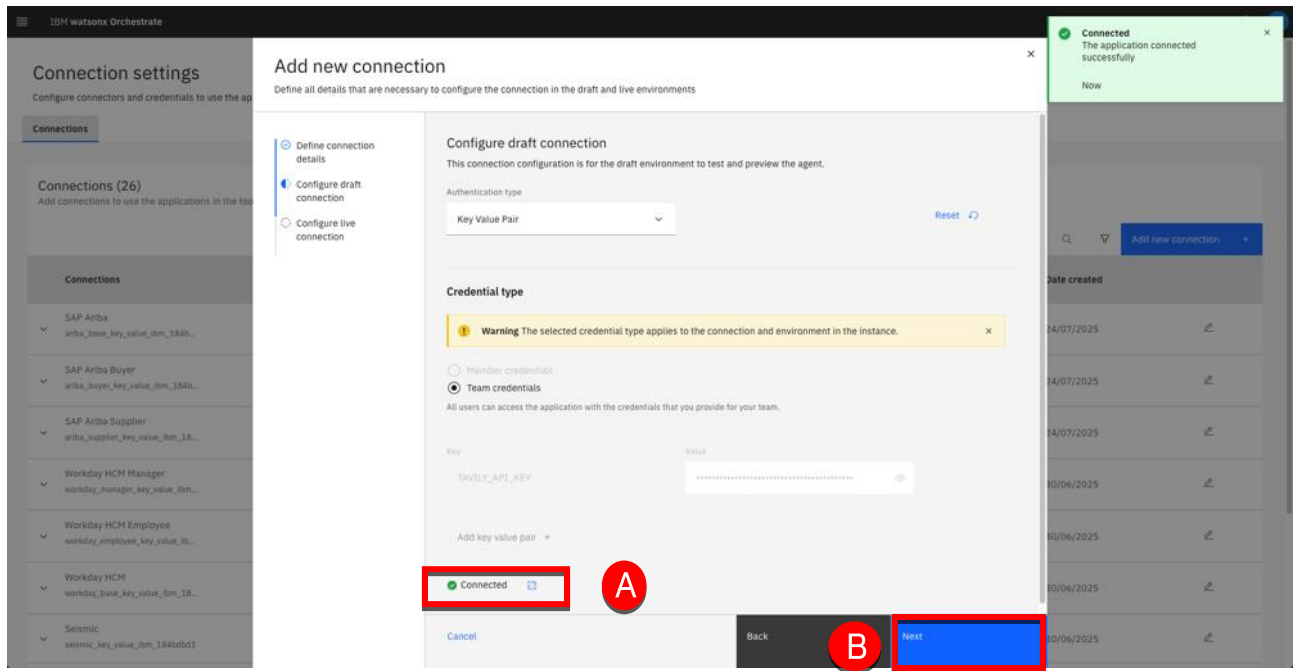
6. Select Key Value Pair (A).



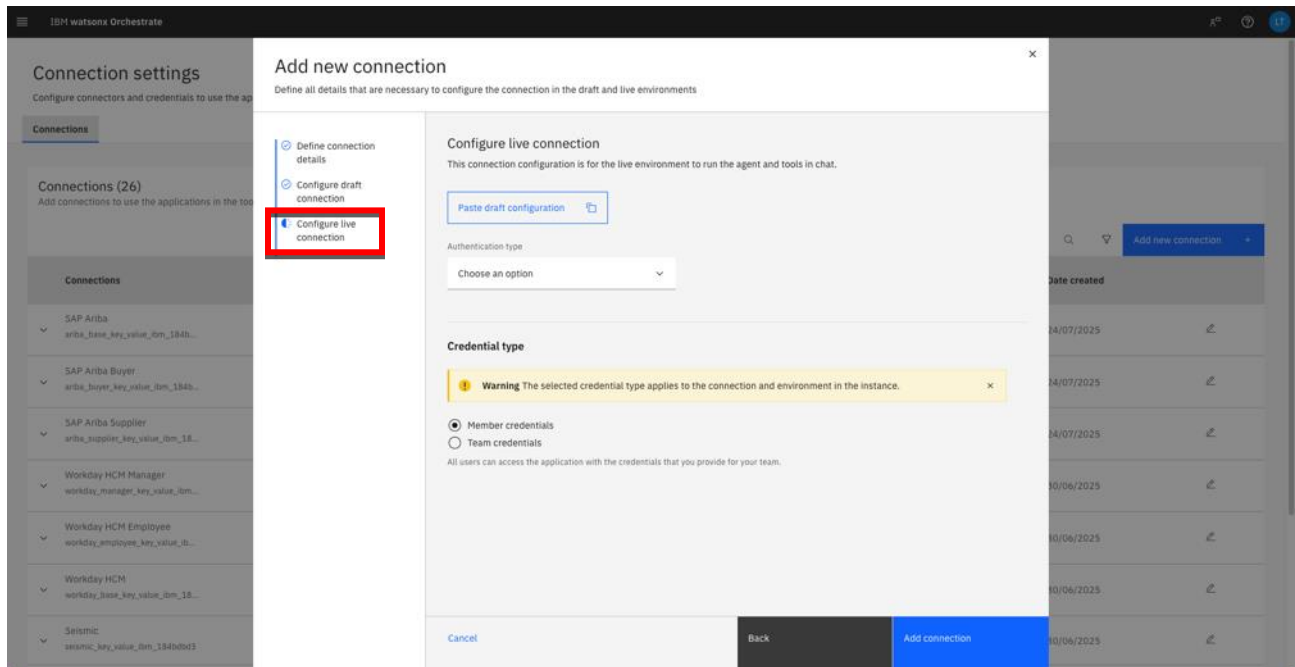
7. Type TAVILY_API_KEY in the **Key** name field (A), copy your Tavily API key from your note in the **Value** field (B), Click **Connect** (C).



8. Check that your draft connection is **Connected** (A), Click **Next** (B).



9. Repeat from Step 5 and add the credentials for the **Live** environment.



10. Click **Add connection** (A) when done.

The screenshot shows the 'Add new connection' dialog in IBM watsonx Orchestrate. The dialog has a sidebar with 'Connections (26)' and a main area for 'Configure live connection'. The 'Authentication type' is set to 'Key Value Pair'. Under 'Credential type', 'Team credentials' is selected. A warning message states: 'Warning The selected credential type applies to the connection and environment in the instance.' The 'Key' field contains 'TAVILY_API_KEY' and the 'Value' field is masked. At the bottom, there are 'Cancel', 'Back', and 'Add connection' buttons. A red box highlights the 'Add connection' button, which is labeled with a red circle 'A'.

11. Check that your connection is now ready (A), both draft and live environments will have access to your credentials in a secured way. Click the top menu (B).

The screenshot shows the 'Connections settings' page in IBM watsonx Orchestrate. The page has a sidebar with 'Connections (27)' and a main area with a table of connections. The table has columns for 'Connections', 'Draft', 'Live', and 'Date created'. The 'Tavily_MCP' connection is highlighted with a red box. In the 'Draft' column, it shows 'Key Value Pair' and 'Team credentials' with a green checkmark. In the 'Live' column, it also shows 'Key Value Pair' and 'Team credentials' with a green checkmark. A red circle 'A' is next to the 'Live' column for this connection. A red circle 'B' is next to the top menu icon.

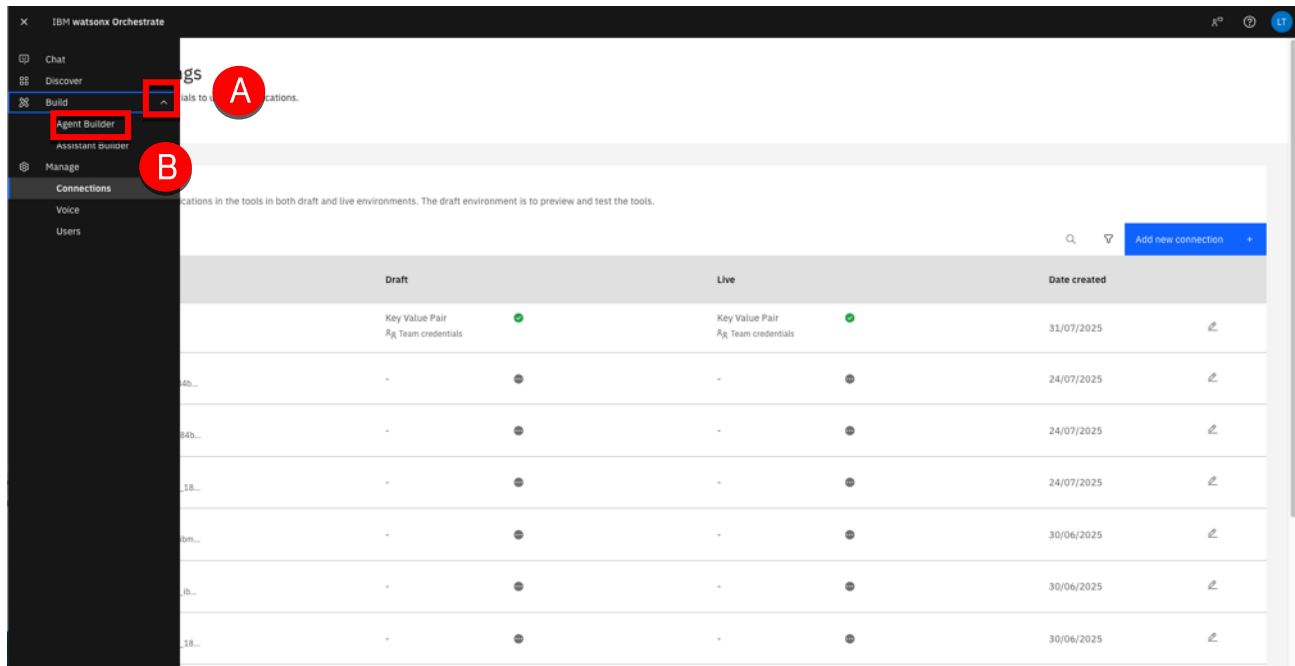
Connections	Draft	Live	Date created
Tavily_MCP	Key Value Pair Team credentials	Key Value Pair Team credentials	31/07/2025
SAP Ariba	-	-	24/07/2025
SAP Ariba Buyer	-	-	24/07/2025
SAP Ariba Supplier	-	-	24/07/2025
Workday HCM Manager	-	-	30/06/2025
Workday HCM Employee	-	-	30/06/2025

Using Local MCP servers to run tools

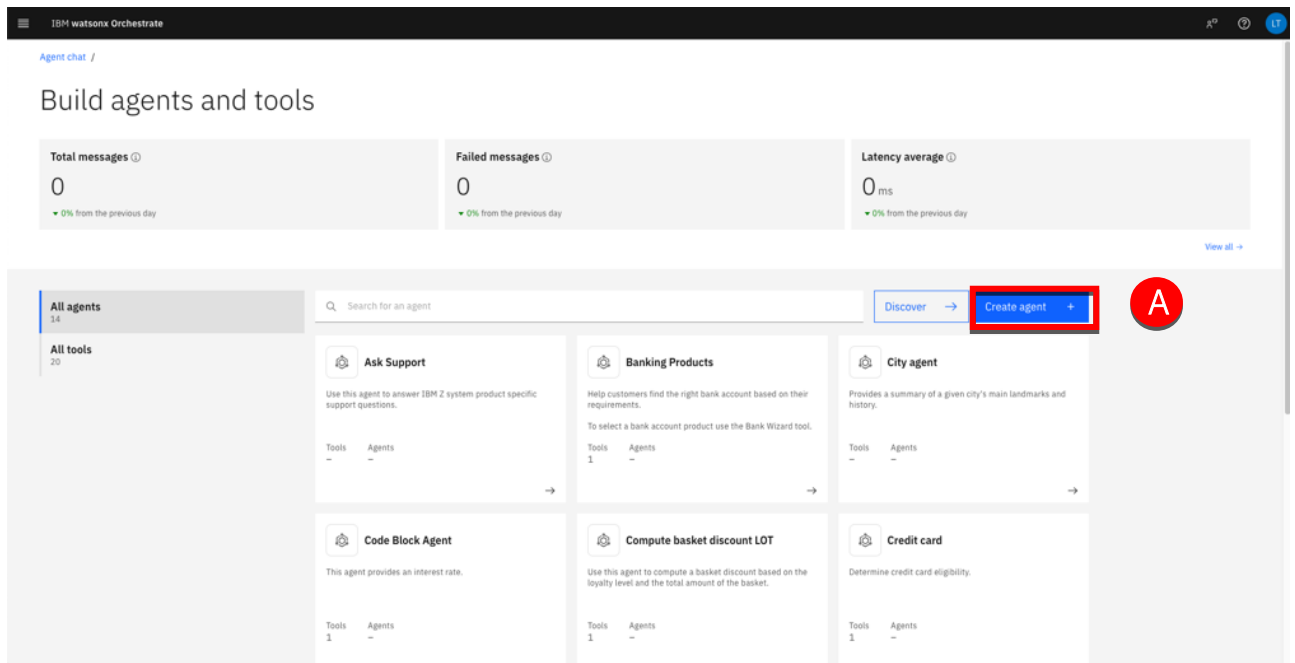
Next, you must create a new agent. This new agent will contain the Tavily search tool you will import from the Tavily MCP server. The connection to the Tavily server will be managed through the connection you have created in the previous step.

Create an agent

1. Expand the **Build** menu (A) then select **Agent Builder** (B).



2. Click **Create agent** (A).



3. Select **Create from scratch** (A), type 'Tavily_search_agent' (B) in the **Name** field. Type 'Use this agent to perform advanced web searches using Tavily. This agent will search over the web based on the input query you will provide.' in the **Description** (C). Click **Create** (D).

The screenshot shows the 'Create an agent' interface in IBM watsonx Orchestrate. The interface has a dark header with the logo and user profile. The main content area is titled 'Create an agent'. There are two tabs: 'Create from scratch' (selected, marked with a red circle A) and 'Create from template'. Below the tabs, there are three input fields: 'Name' (marked with a red circle B), 'Description' (marked with a red circle C), and a 'Create' button (marked with a red circle D). The 'Name' field contains the text 'Tavily_search_agent_LOT2508'. The 'Description' field contains the text 'Use this agent to perform advanced web searches using Tavily. This agent will search over the web based on the input query you will provide.' The 'Create' button is a blue button with the text 'Create'.

IBM watsonx Orchestrate

Create an agent

A

Create from scratch
Build your custom agent step by step to create a custom solution.

Create from template
Browse the catalog and use the attributes of another agent as a template to create your agent.

B

Name
Tavily_search_agent_LOT2508

C

Description
Use this agent to perform advanced web searches using Tavily. This agent will search over the web based on the input query you will provide.

What makes a good description? [?](#)

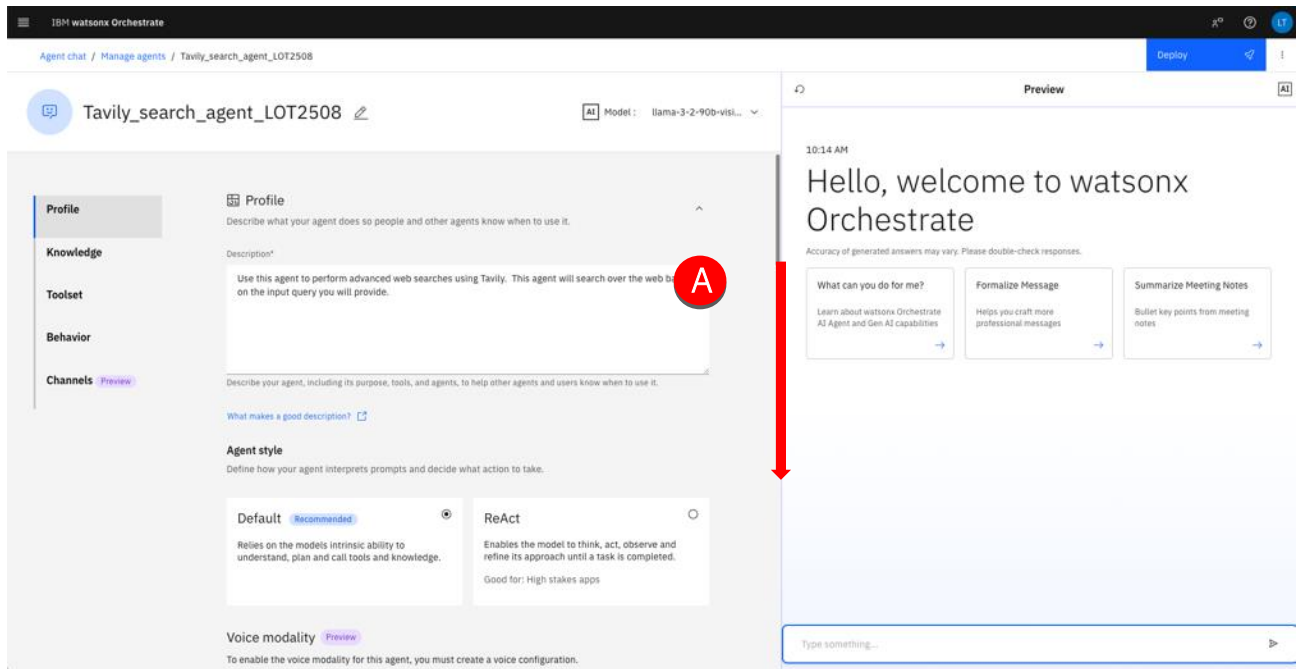
Cancel

D

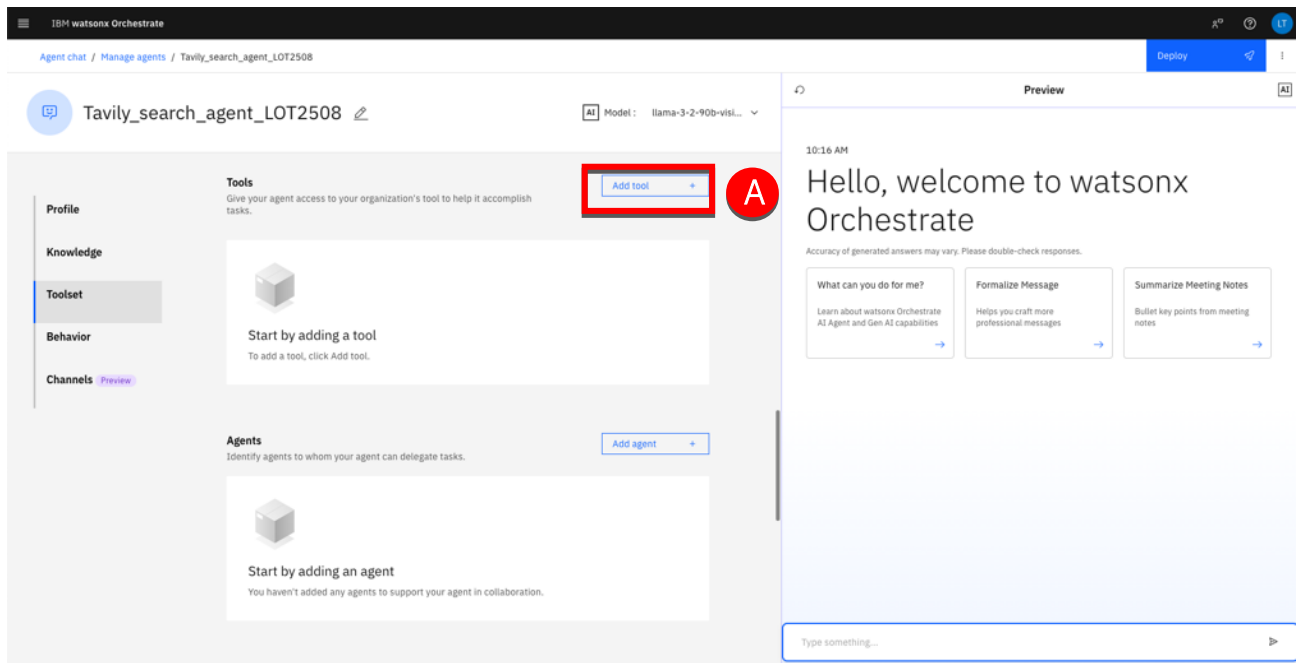
Create

Configure an MCP Server

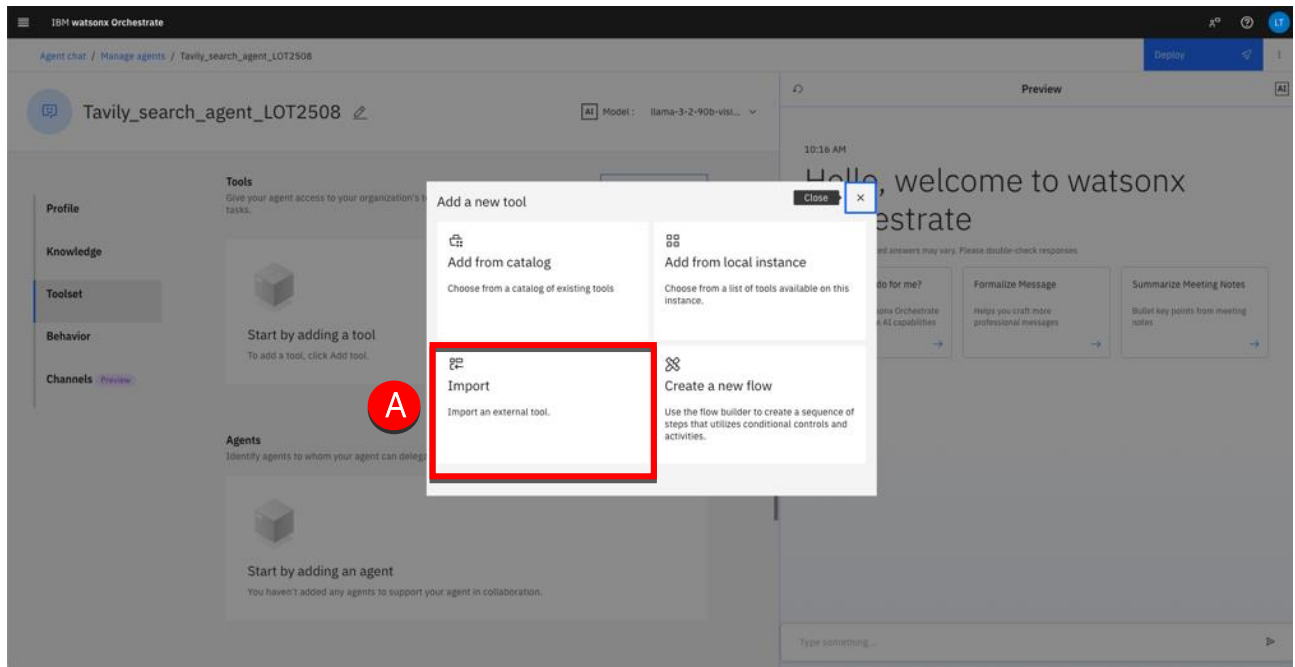
1. Scroll down to the **toolset** section of your agent (A).



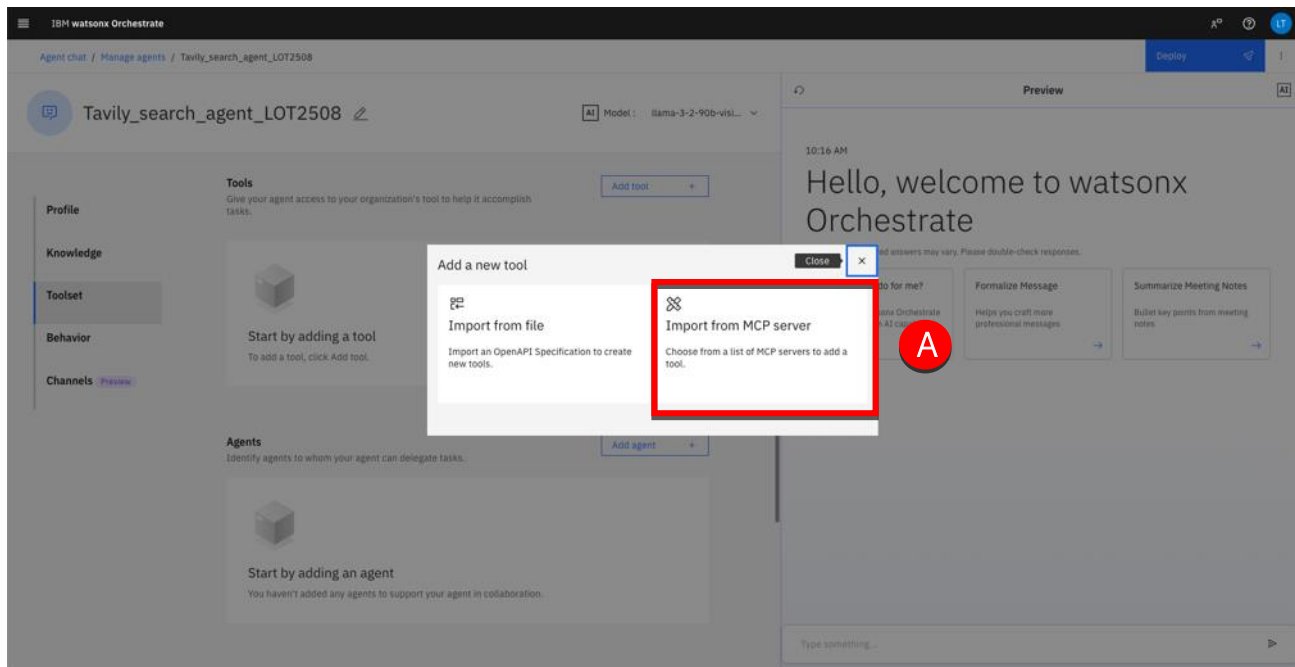
2. Click **Add tool** (A).



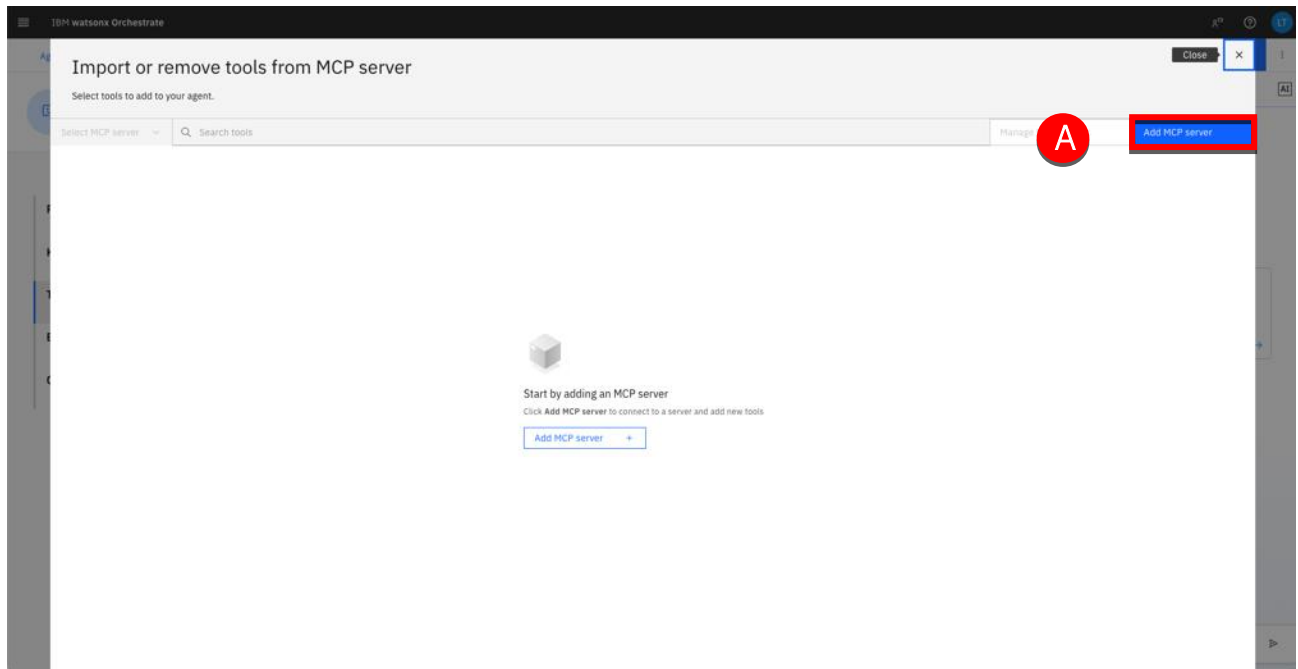
3. Click **Import** (A).



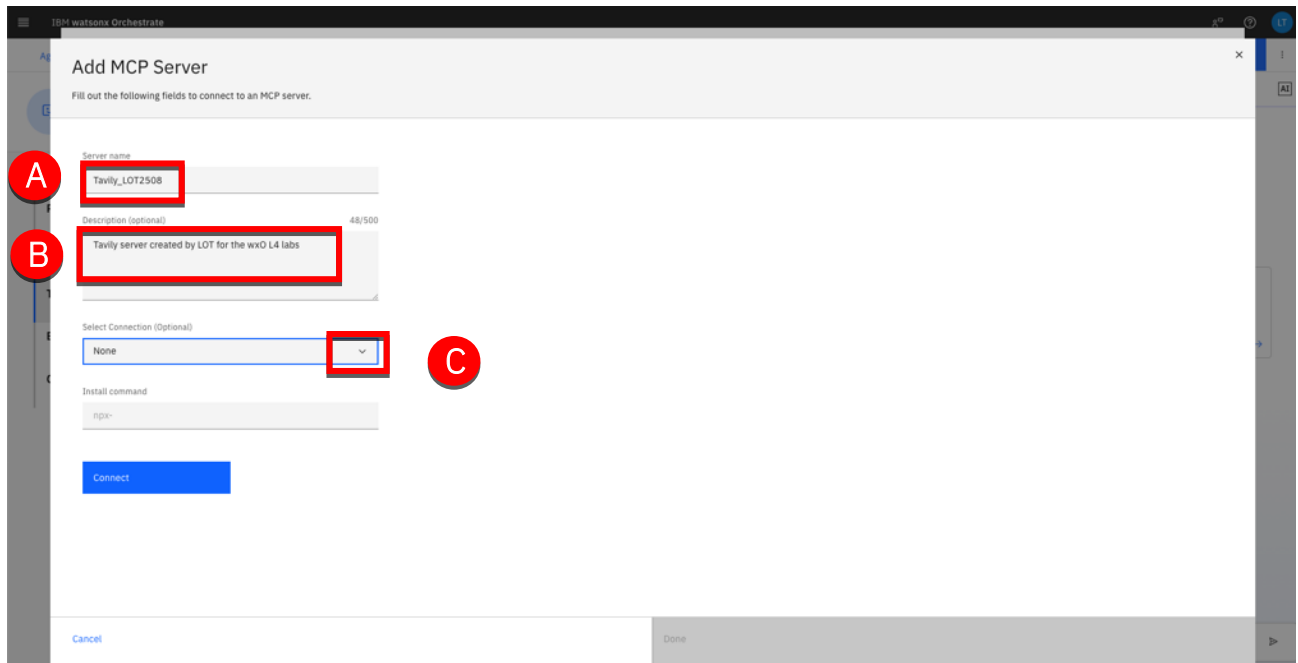
4. Click **Import from an MCP server** (A).



5. Click **Add MCP server** (A).



6. Type 'Tavily' in the **Server name** (A), type 'Tavily server created for the wx0 L4 labs' in the **Description** (B), expand the **Select Connection** drop down (C).



7. Select your **Tavily_MCP** from the list (A).

IBM watsonx Orchestrate

Add MCP Server

Fill out the following fields to connect to an MCP server.

Server name
Tavily_LOT2508

Description (optional) 48/500
Tavily server created by LOT for the wxO L4 labs

Select Connection (Optional)

- TAVILY_MCP** (A)
- SAP Ariba Buyer
- SAP Ariba Supplier
- Workday HCM Manager
- Workday HCM Employee
- Workday HCM
- Seismic
- SAP SuccessFactors URL
- Coupa
- Oracle URL
- Slack
- Google Drive
- SAP SuccessFactors
- MS Outlook
- Slack URL
- ServiceNow
- ServiceNow URL
- Salesforce
- Box

Done

8. Type 'npx -y tavily-mcp@0.1.3' in the command field (A) then click **Connect** (B).

IBM watsonx Orchestrate

Add MCP Server

Fill out the following fields to connect to an MCP server.

Server name
Tavily_LOT2508

Description (optional) 48/500
Tavily server created by LOT for the wxO L4 labs

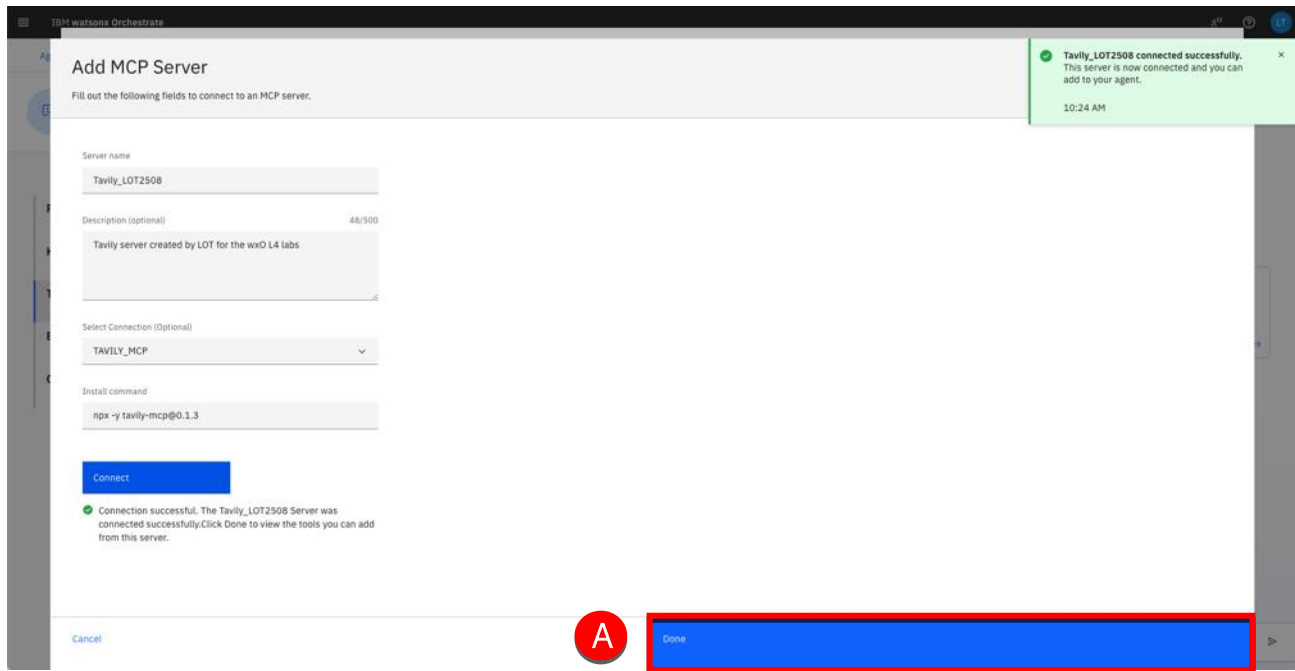
Select Connection (Optional)
TAVILY_MCP

Install command
npx -y tavily-mcp@0.1.3 (A)

Connect (B)

Cancel Done

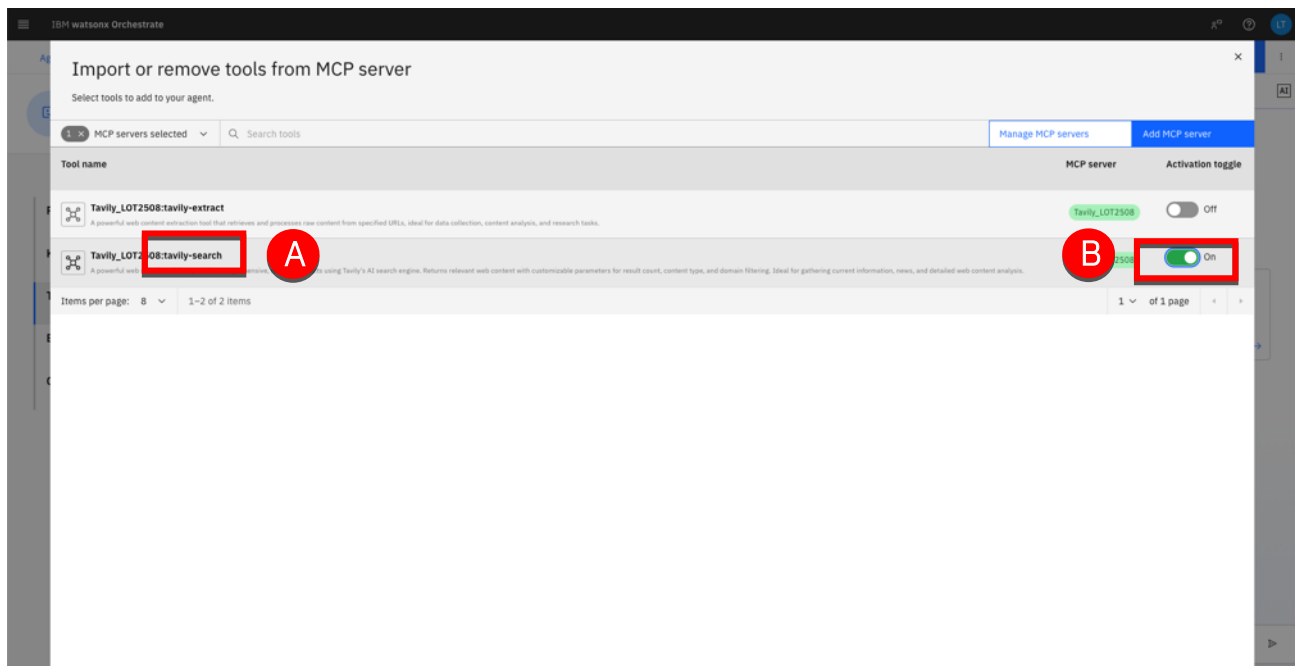
9. Click **Done** (A).



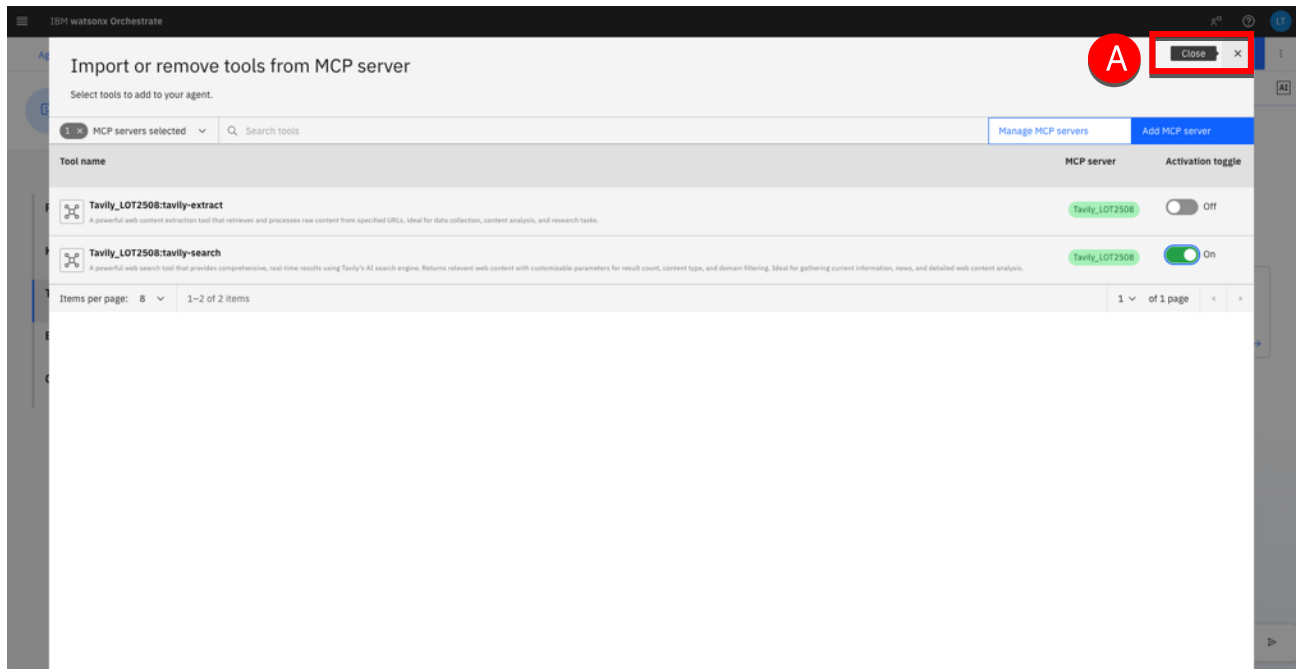
Select the tools to import from the MCP server

Now that your server is imported and you are connected to the server, you can add the tools that are exposed to your agent.

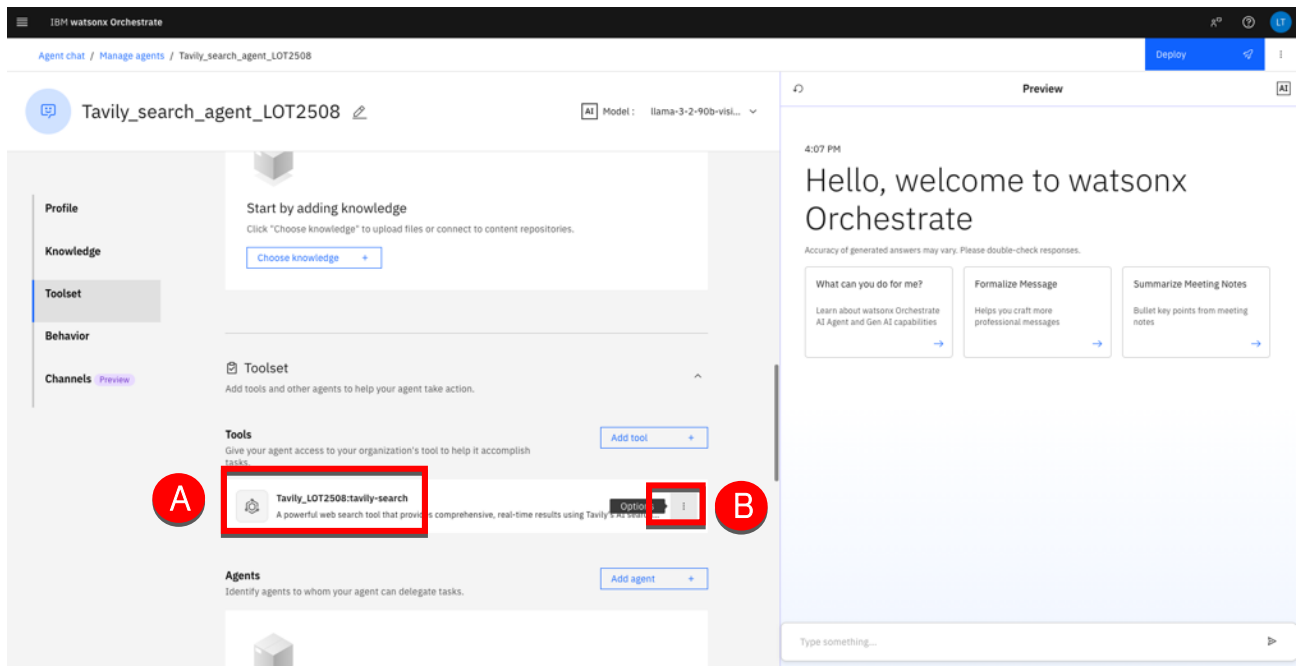
1. Identify the **tavily_search** tool (A) then switch its **Activation toggle** to On (B).



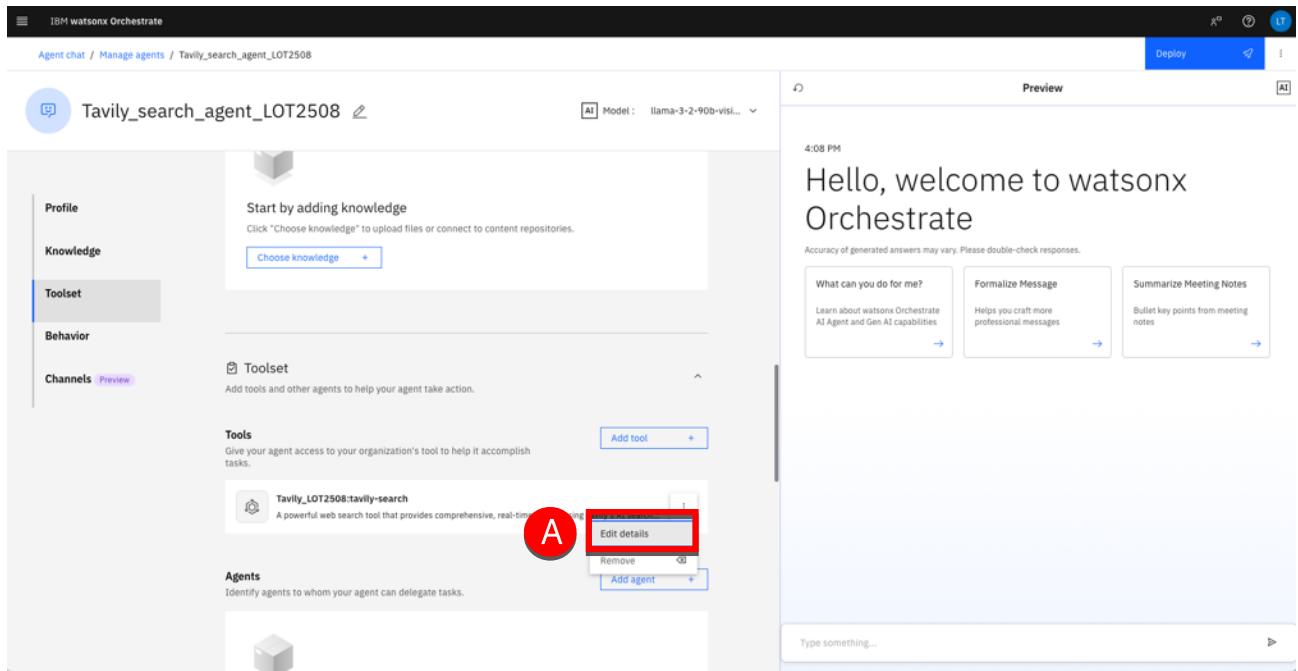
- Click X to close the view (A).



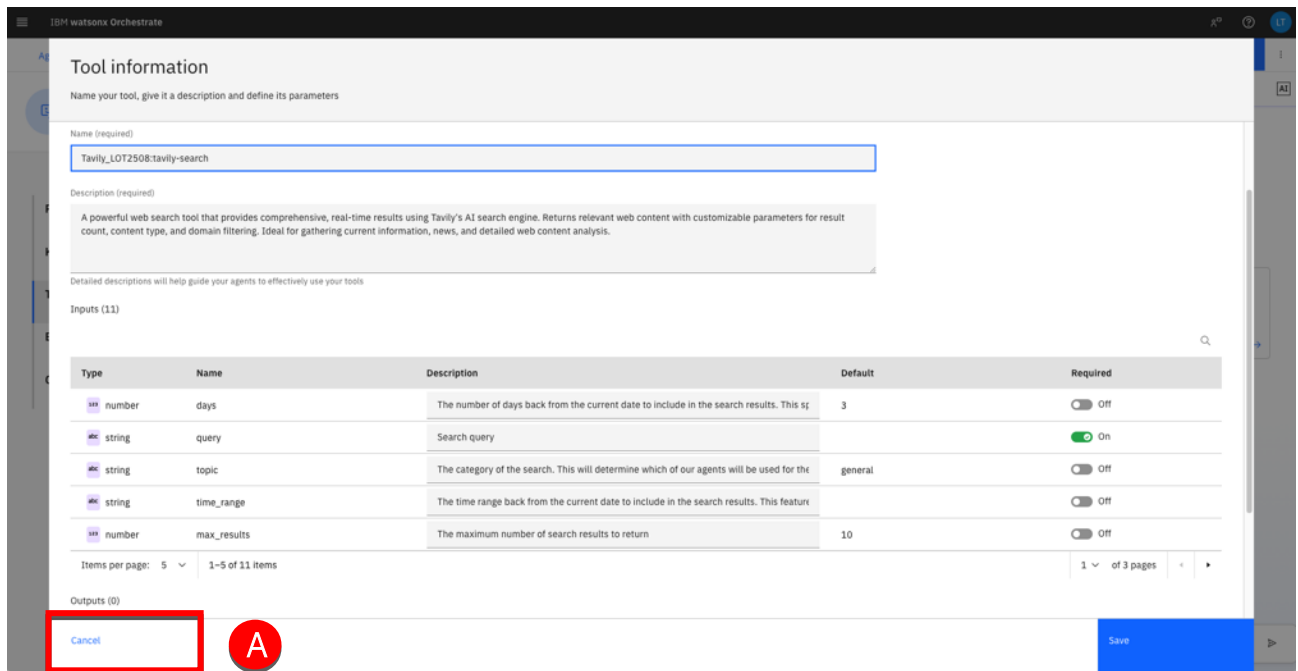
- Note that the Tavily search tool is now added to your own tools (A). Tavily is an advanced web search tool. More parameters can be specified in addition to the default search query. Click the option menu icon (B).



4. Click **Edit details** (A).



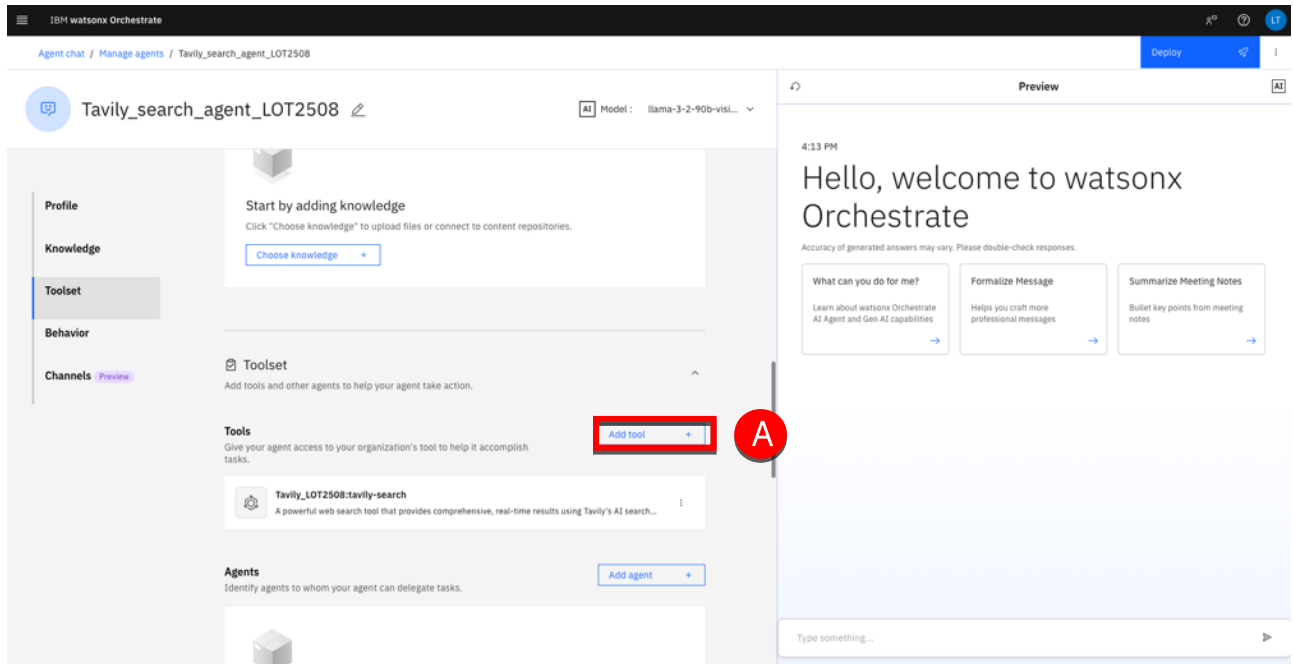
5. By default, only the search query is required to launch the tool. But you can activate more parameters to specify the topic of the search or limit the search to the recent content specifying the number of days since the retrieved content was published. Let's keep with the default query for now. Click **Cancel** (A).



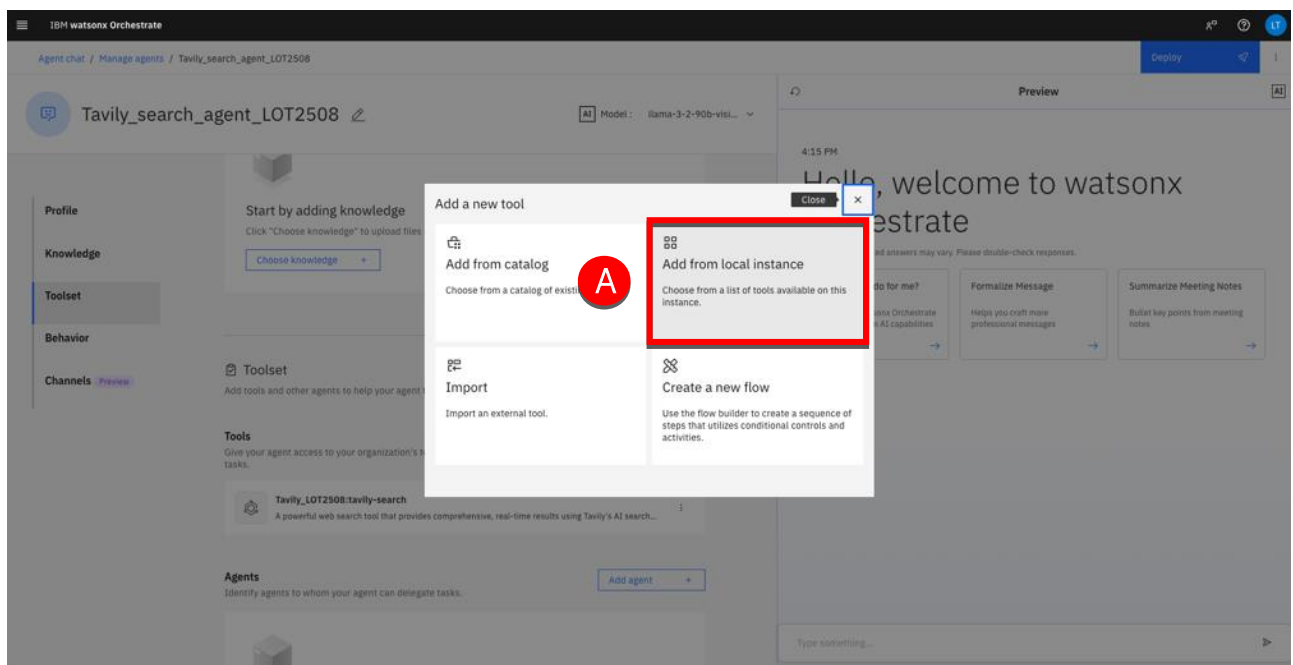
Adding more tools from your local MCP servers.

MCP servers are created just once when adding the first tool in your watsonx Orchestrate instance. Once the specified import command executed, you can add more tools (if any) from this MCP server. Let's see how.

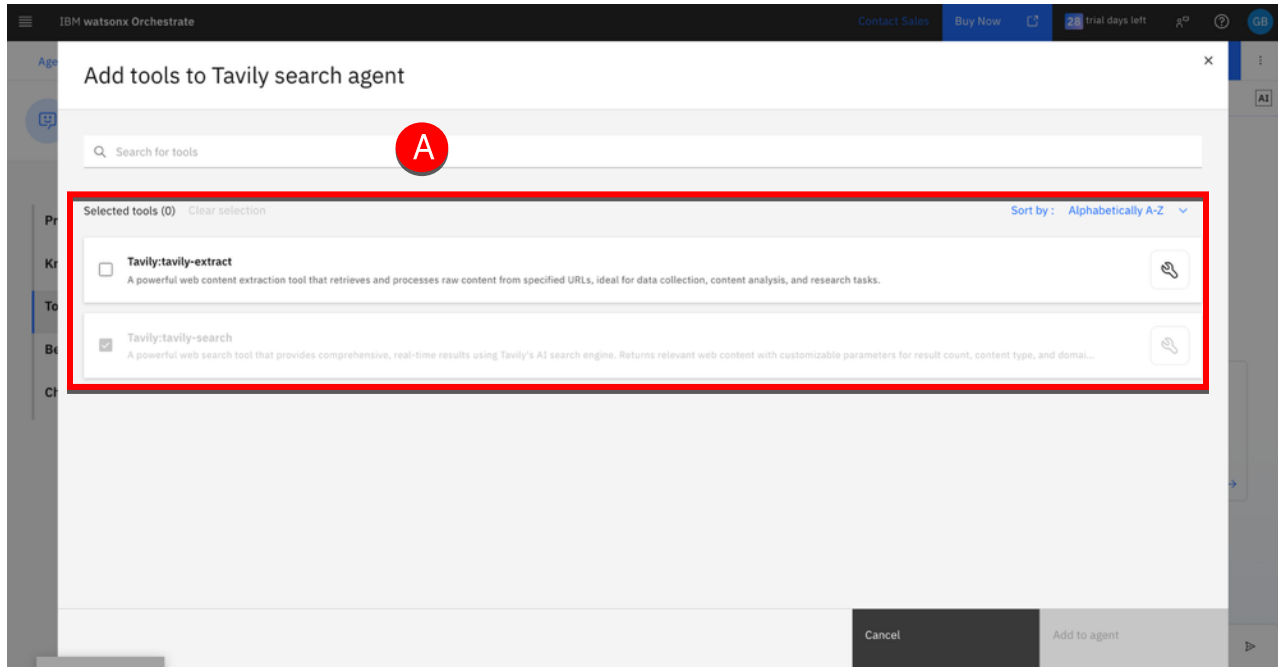
1. Click **Add tool** (A).



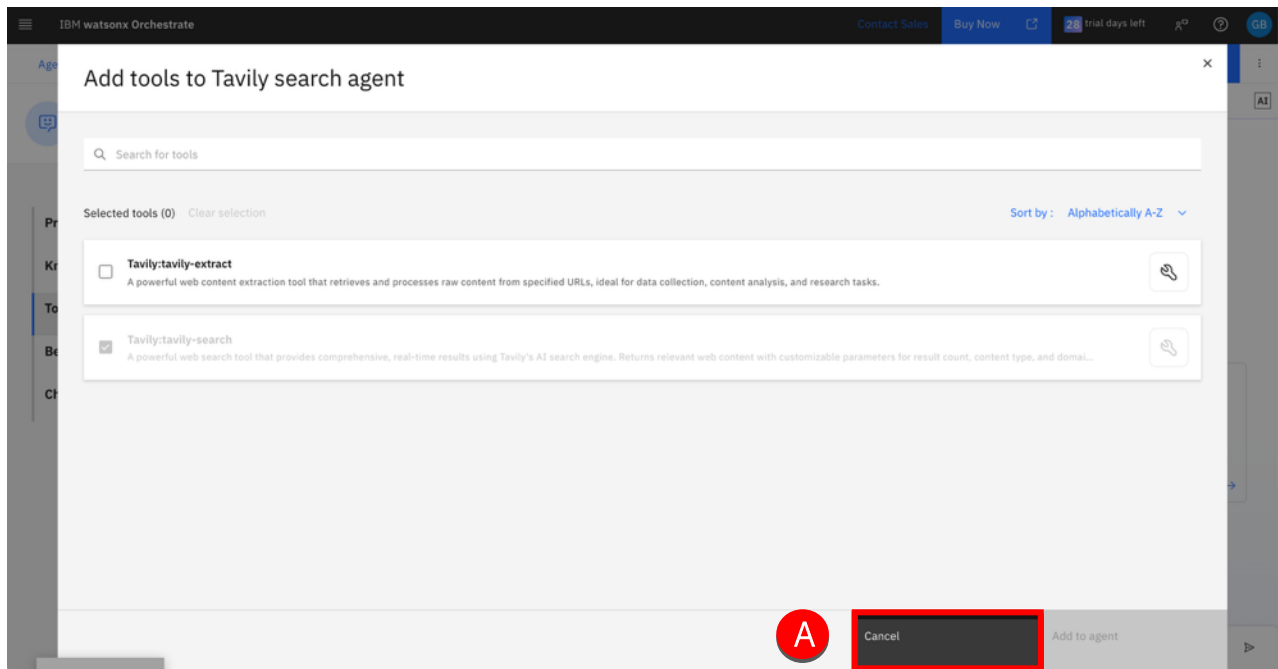
2. Select **Add from a local instance** (A).



3. If you have created other tools they will be listed here (A). If you are using a new tenant, only the Tavily tools will be listed.



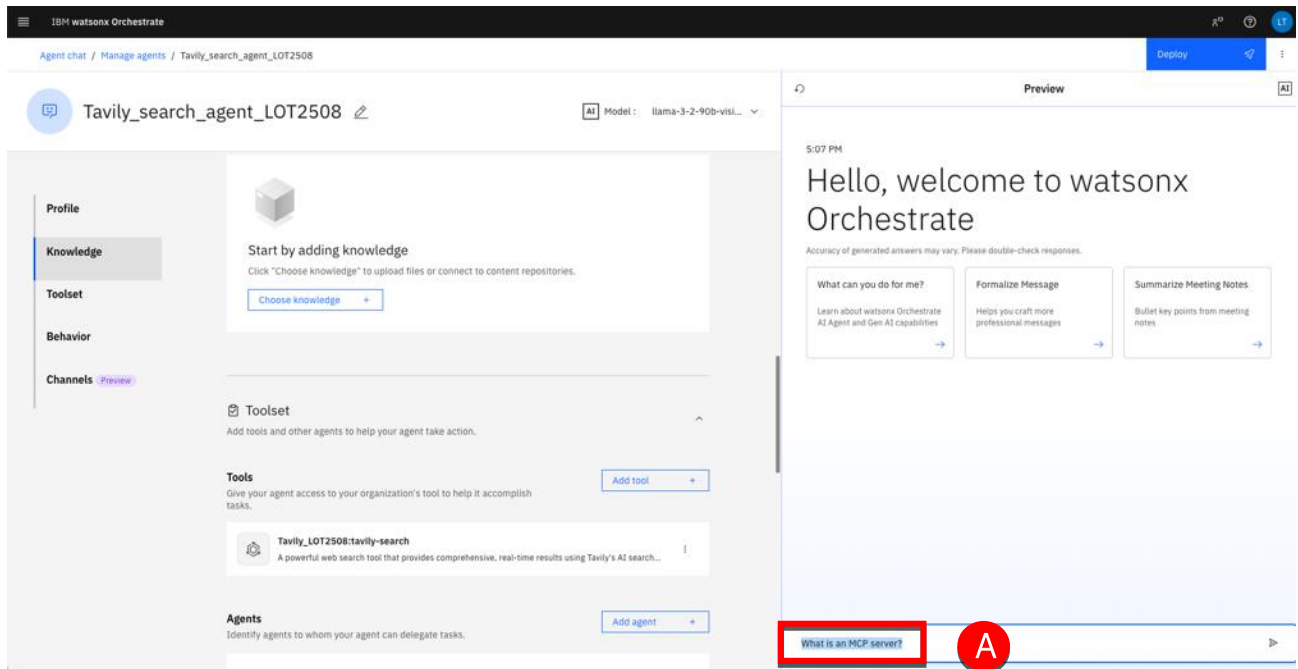
4. The Tavily server exposes 2 tools: The Tavily-extract and Tavily-search. The search that you have already added to your agent is grayed out. From here, you can decide to add more tools at any time. We will just use the Search tool in this lab. Click **Cancel** (A).



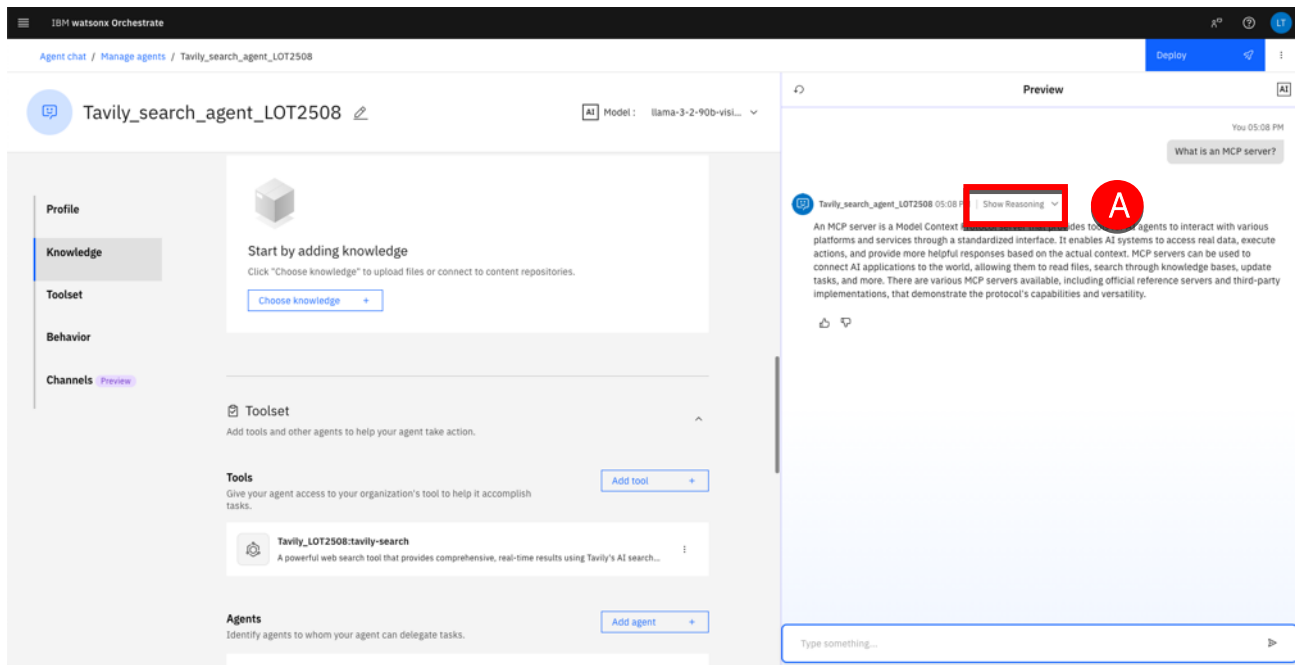
Preview the tool within your agent

Your tool is now ready to be used in your agent. Let's use watsonx Orchestrate preview to play with it.

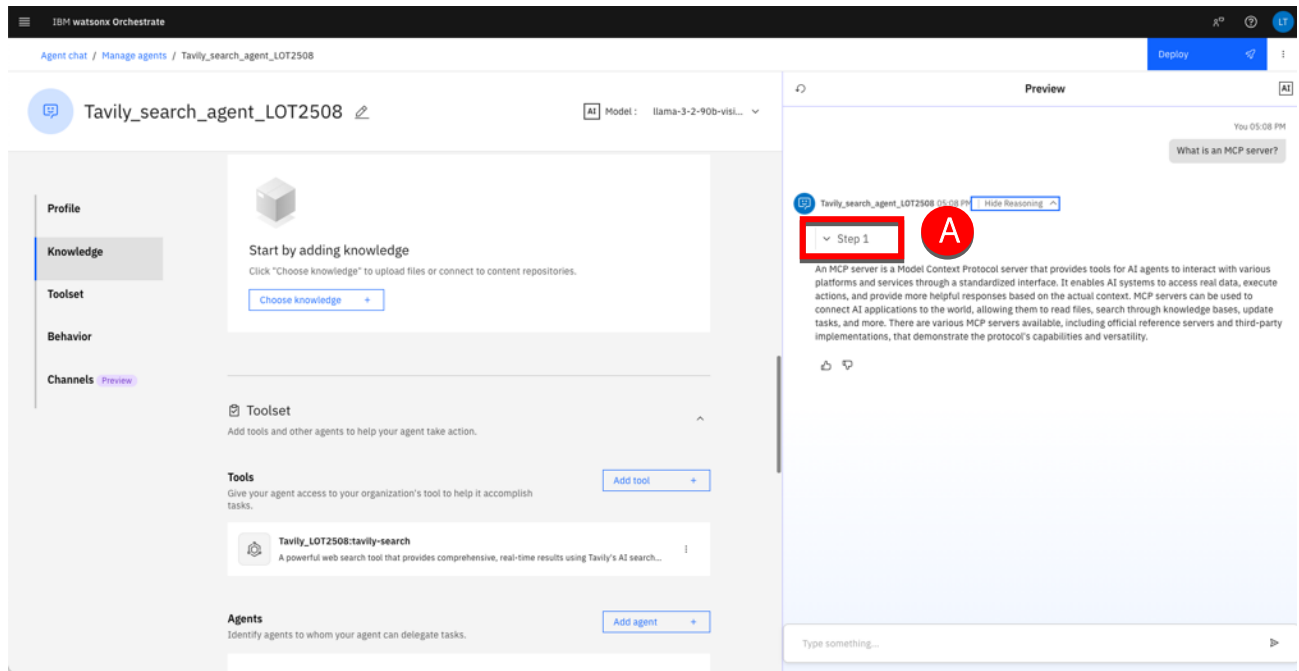
1. Type 'What is an MCP server?' in the chat area (A).



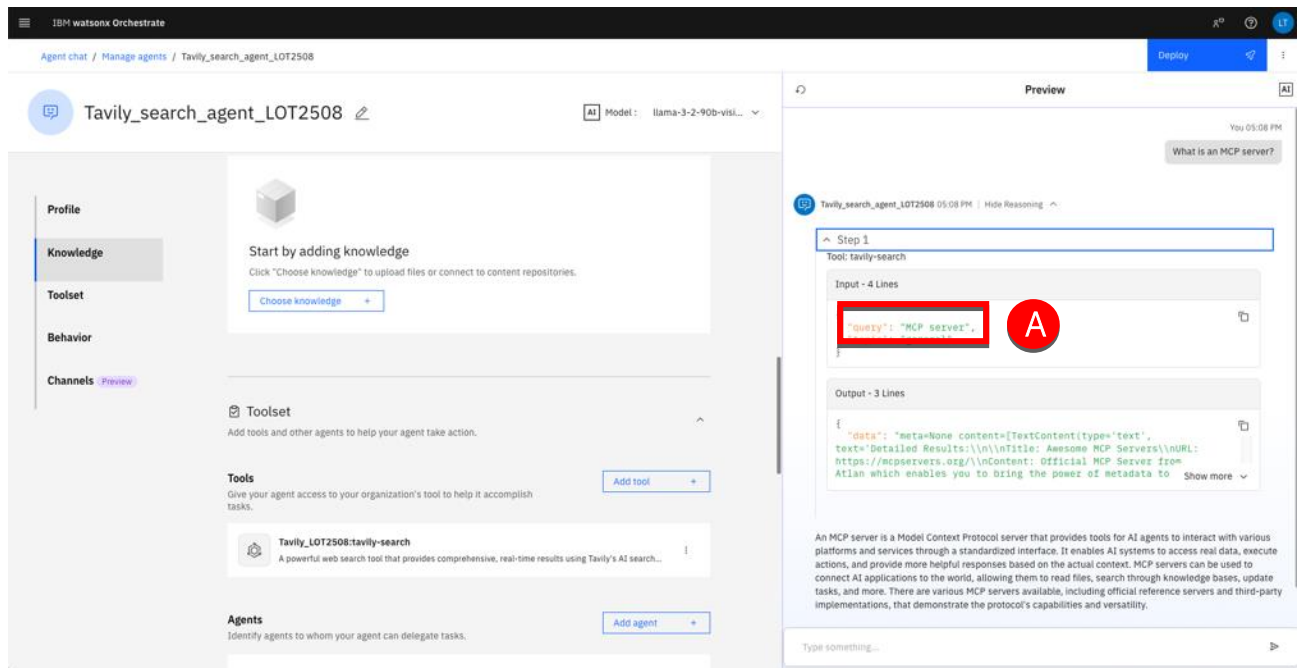
2. Expand Show reasoning (A).



3. Expand Step 1 (A).



4. Note the query the agent extracted from your input query and how it was used to invoke the Tavily search engine (A). By default, the topic was set to 'General' as haven't specified this input parameter as required in the tool settings in the previous steps.



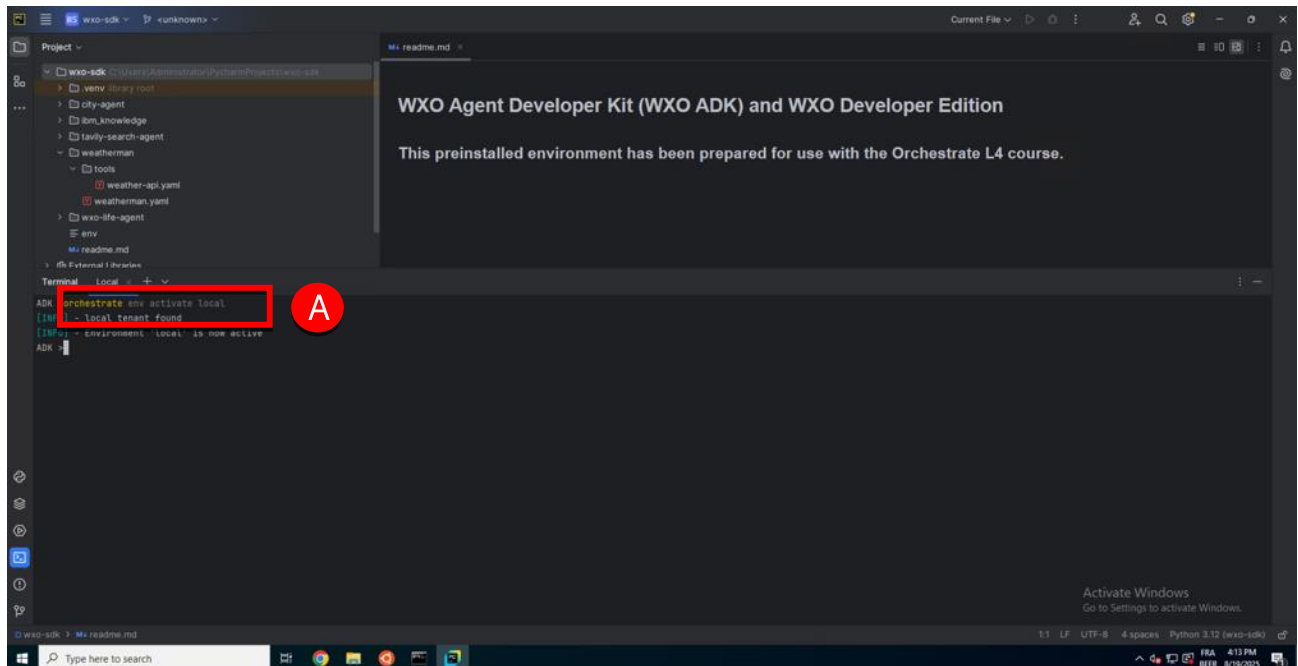
PART 2: Import MCP tools using watsonx Orchestrate Developer Edition (ADK)

In this first part, you will learn how to work with MCP tools using the watsonx Orchestrate Developer Edition using the ADK commands.

Open your ADK environment

1. Open your ADK terminal as described in the preparation guide. Refer to the [Reset your environment appendix](#) if your server is not started.
2. Enter the following command to make sure you are running on the local environment (A).

```
orchestrate env activate local
```



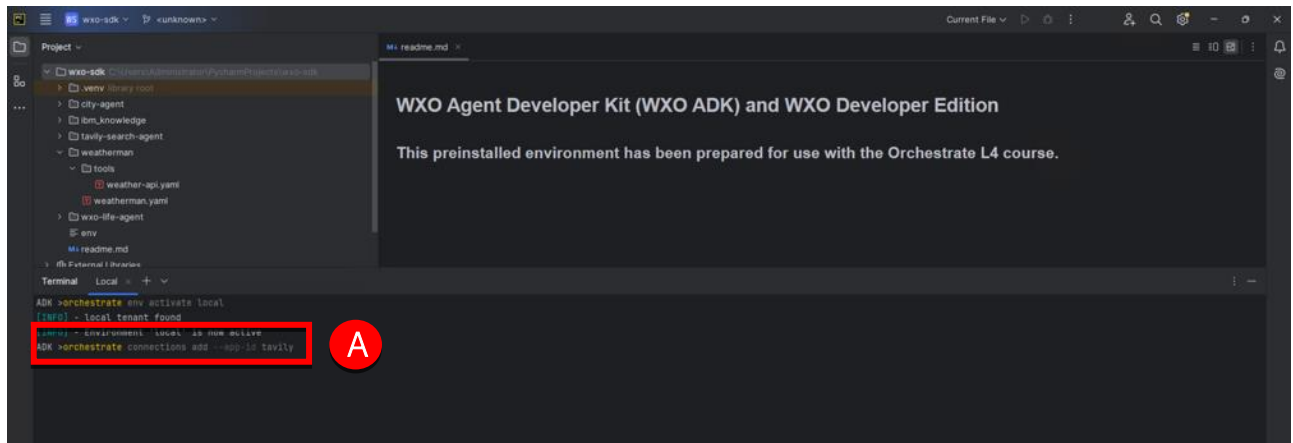
Create a Tavily connection

Like you have performed in part 1, a Tavily connection must be created to connect watsonx Orchestrate to the Tavily server you are using.

Create the connection

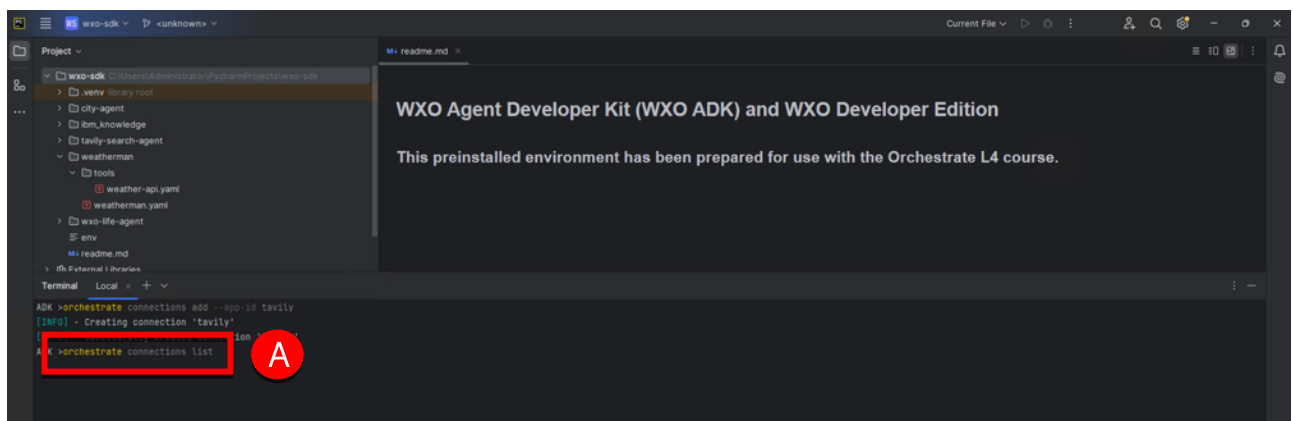
1. Type the following command to create a connection with an application id called tavily

```
orchestrate connections add --app-id Tavily
```

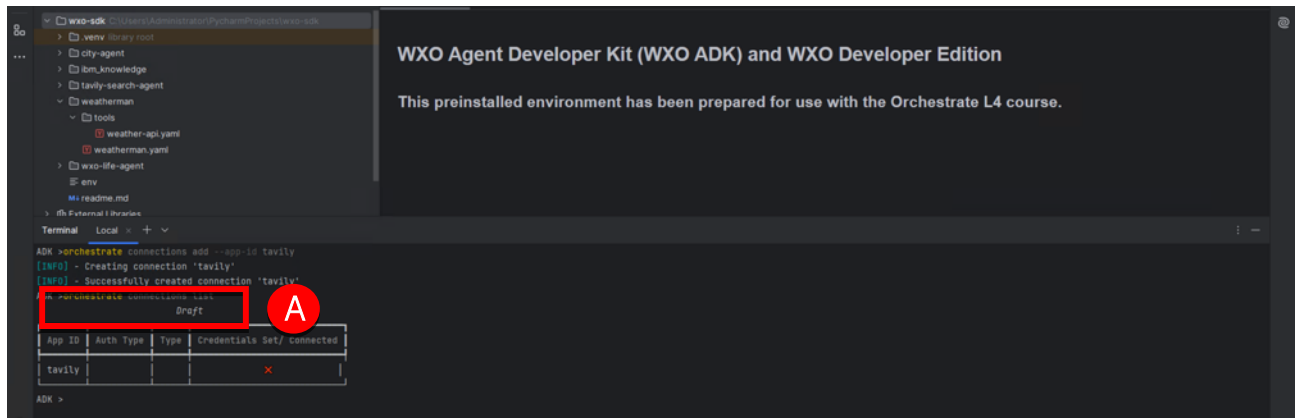


2. After receiving the success confirmation message, type the following command to list the existing connections on your local environment (A)

```
orchestrate connections list
```



3. Note your Tavily connection has been created (A). As you are running on watsonx Orchestrate Developer Edition, only Draft environments are supported. This is why no live connection is listed.



Note: You can switch to a SaaS environment to create the MCP tools if you want to work with Draft and Live environments. To learn how to switch to a SaaS environment, please refer to the ADK – Using ADK with SaaS lab. You can then repeat the following instructions replacing Draft with live.

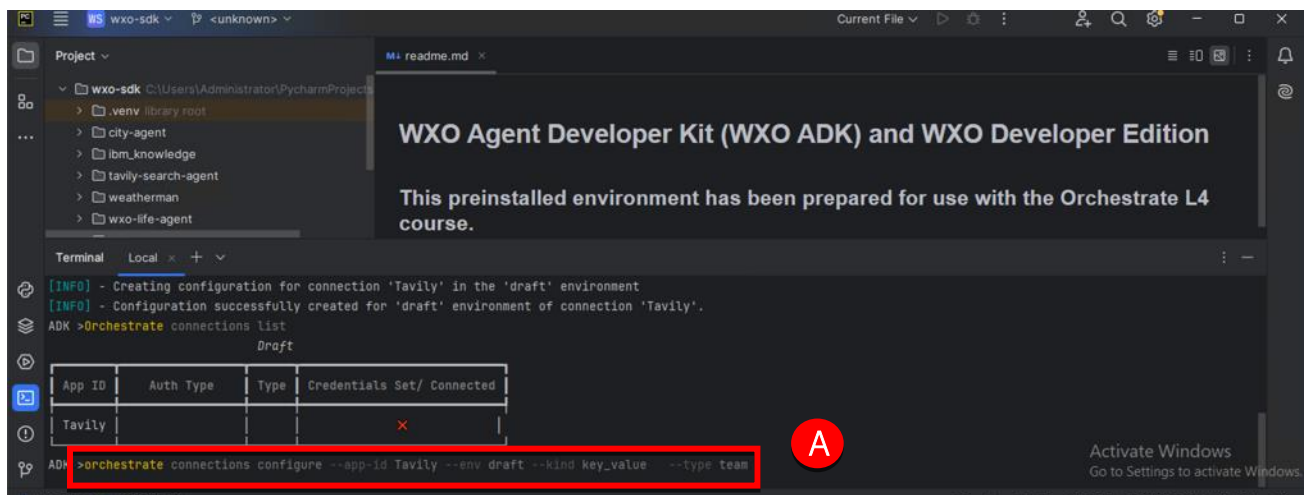
Configure the environments parameters

Next you will configure the connection parameters as you did in the previous part through the UI:

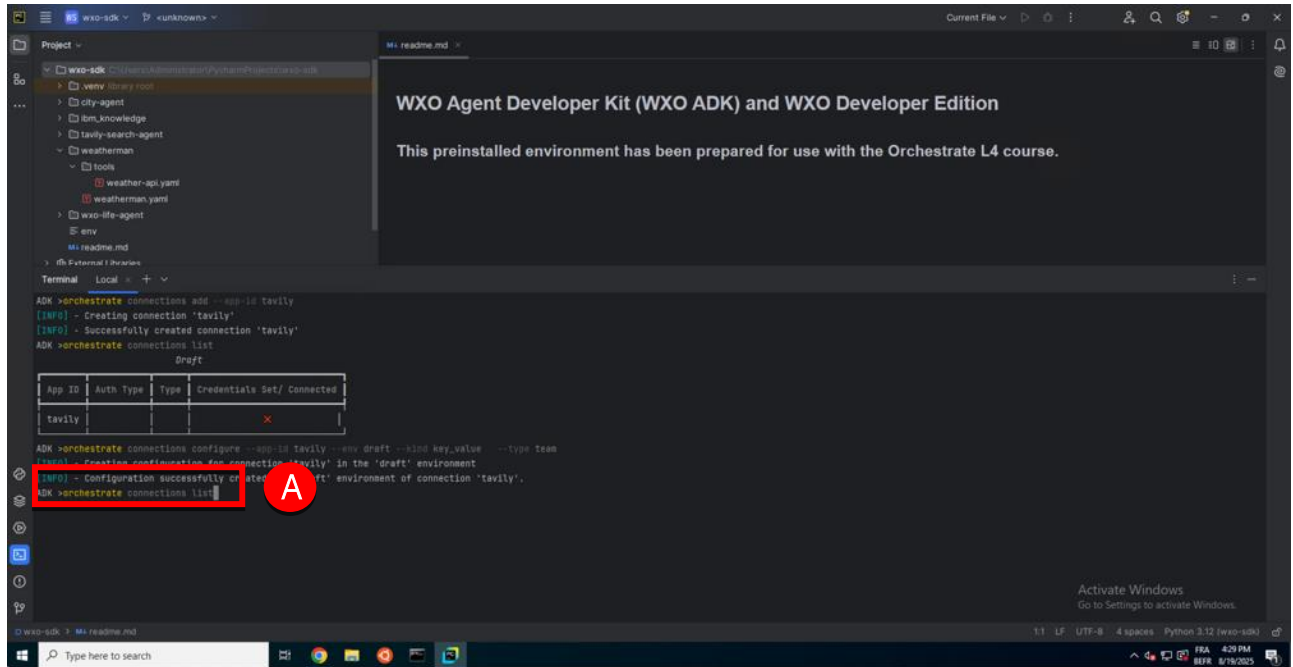
- You will configure the **draft** env
- It will use **key_value** pairs
- Credentials will be used for each **team** members

1. Enter the following command to configure the connection (A).

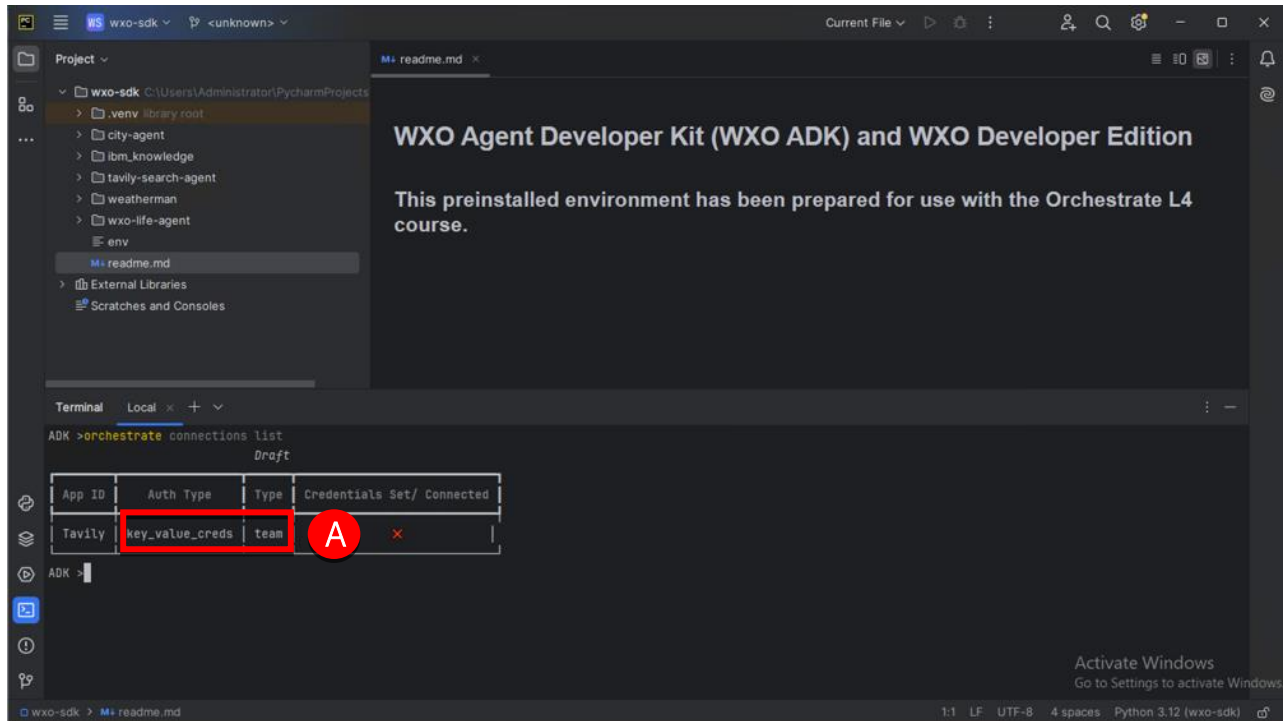
```
orchestrate connections configure --app-id Tavily --env draft --kind key_value --type team
```



- After receiving the success notification message type 'orchestrate connections list' again (A).



- Note that the connection parameters are now set (A). Next you will provide the credentials to connect your Tavily instance.

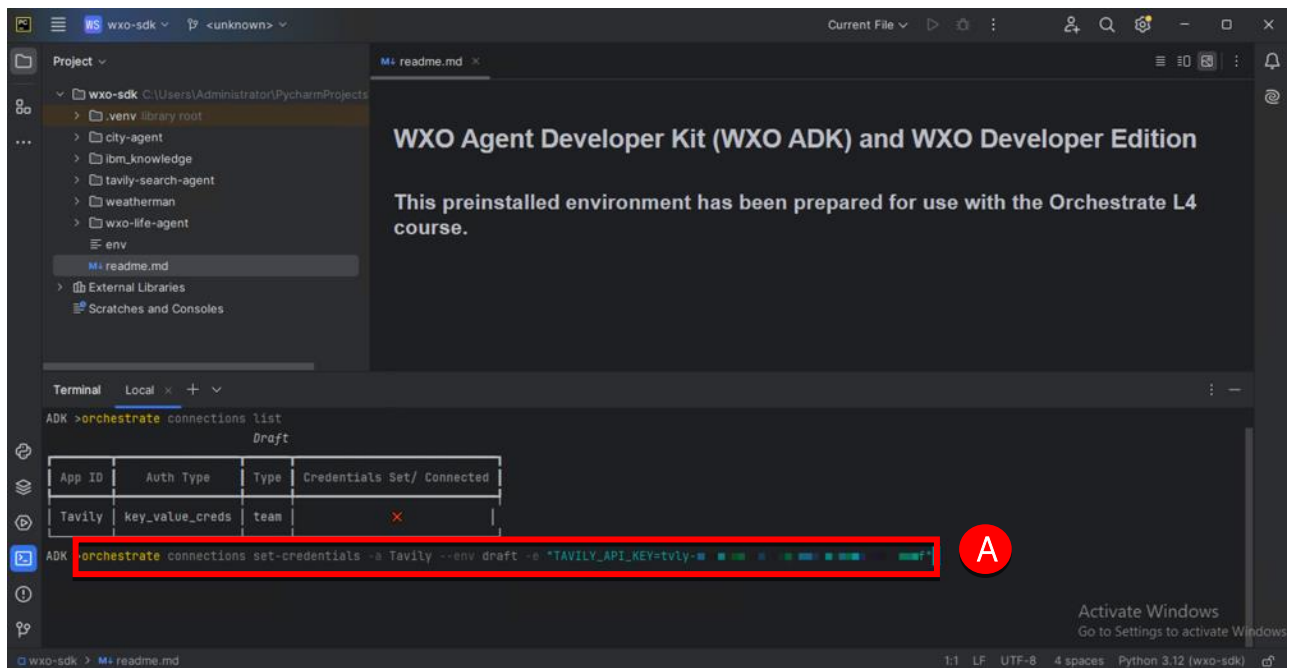


Specify the credentials to connect your tavily server

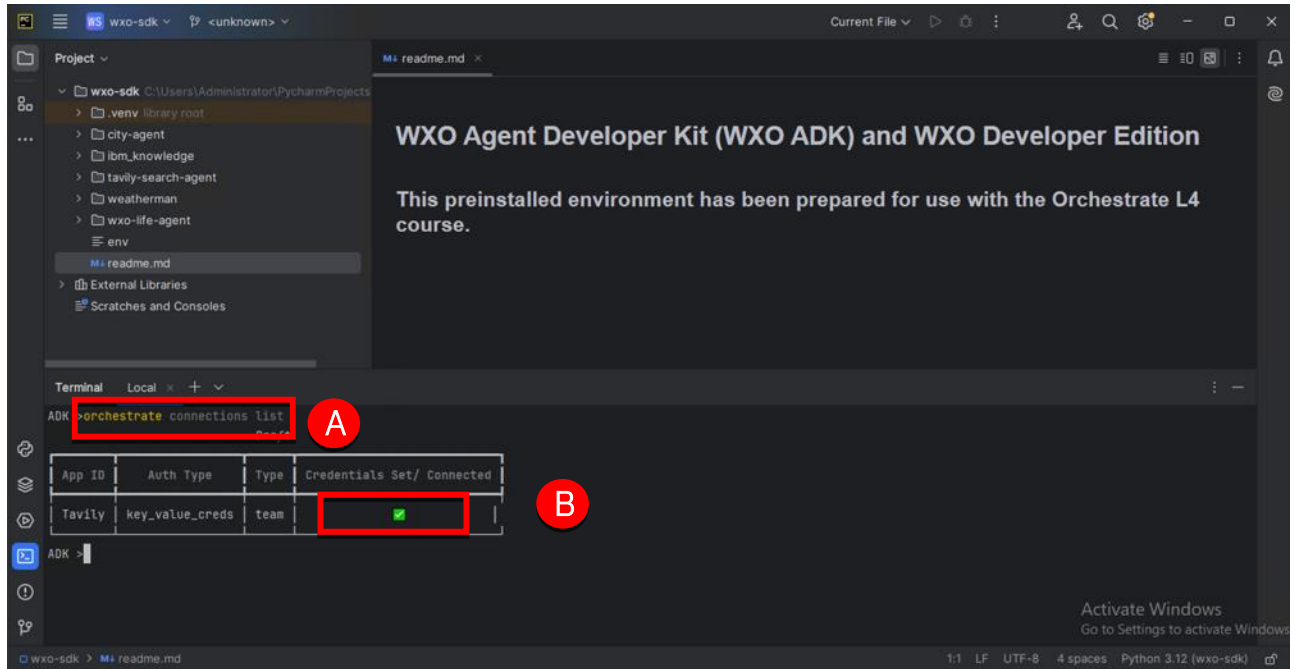
The authentication method is to use key_value pairs. You have already configured your Tavily server in the Preparation guide. You must now provide this key to the Tavily connection you have created.

4. Enter the following command replacing <your Tavily api key> with your own api key collected in item 11 of the prep guide (A)

```
orchestrate connections set-credentials -a Tavily --env draft -e  
"TAVILY_API_KEY=<your tavily api key>"
```



5. After receiving the success notification message, type again 'orchestrate connections list' (A) and note that the connection is now completely configured and connected (B).

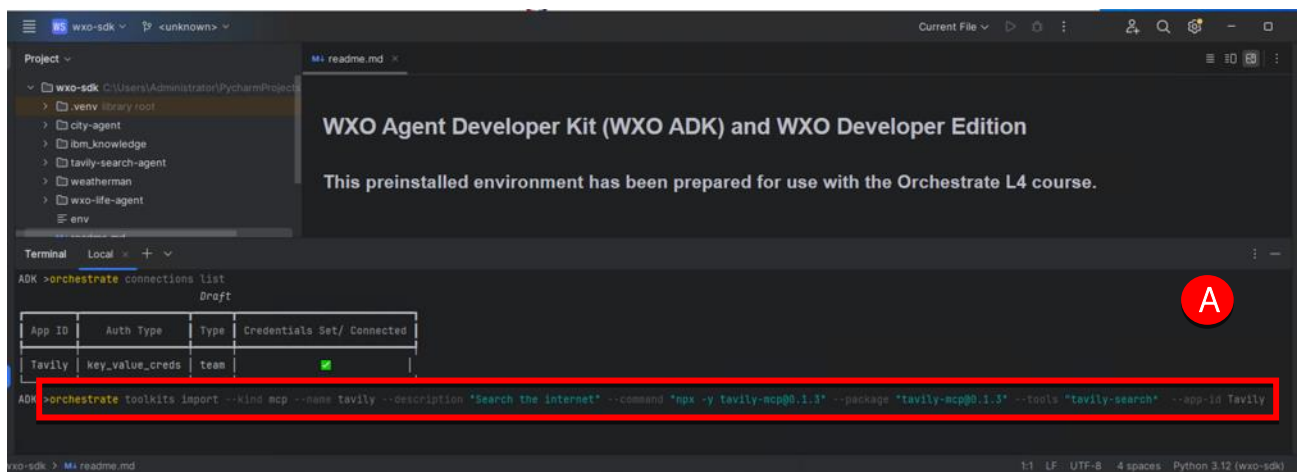


Import Tavily tools in a Tavily toolkit

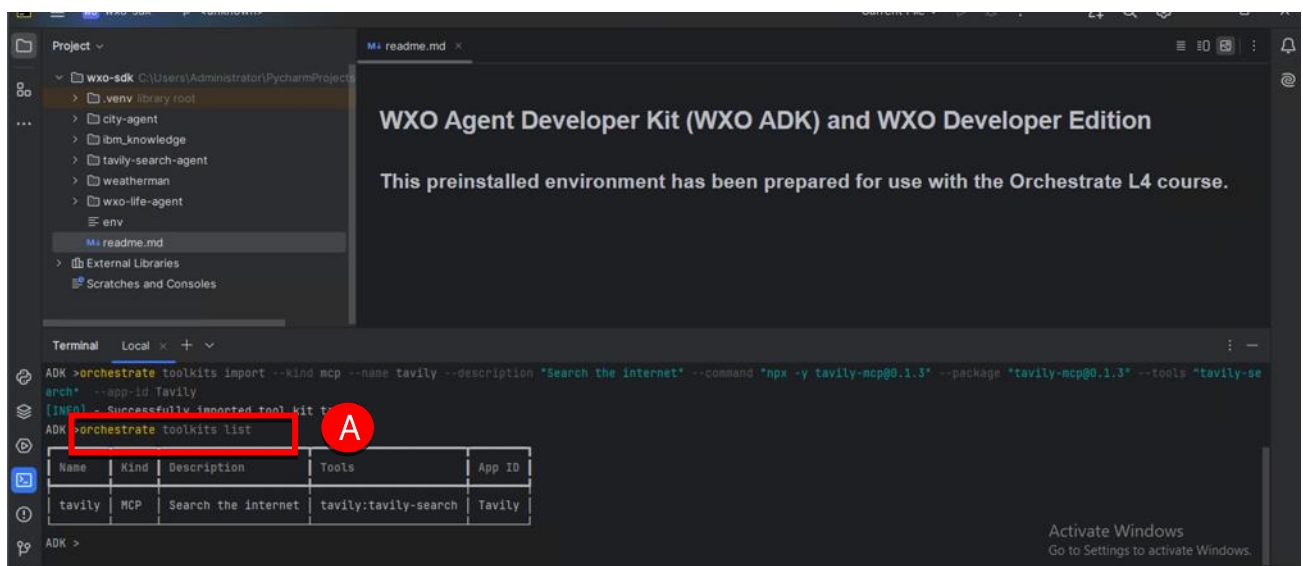
Next you will import **mcp kind** search tool. The tool to import from Tavily is “**Tavily-search**” and the **npx -y tavily-mcp@0.1.3** command will be used to import the tool locally. The tool description will be “**Search the internet**” in this lab.

1. Type the following command to import the Tavily-search tool in your Tavily toolkit.

```
orchestrate toolkits import --kind mcp --name tavily --description "Search the internet" --command "npx -y tavily-mcp@0.1.3" --package "tavily-mcp@0.1.3" --tools "tavily-search" --app-id Tavily
```



2. Wait for the toolkit creation to be performed then type 'orchestrate toolkits list' (A).

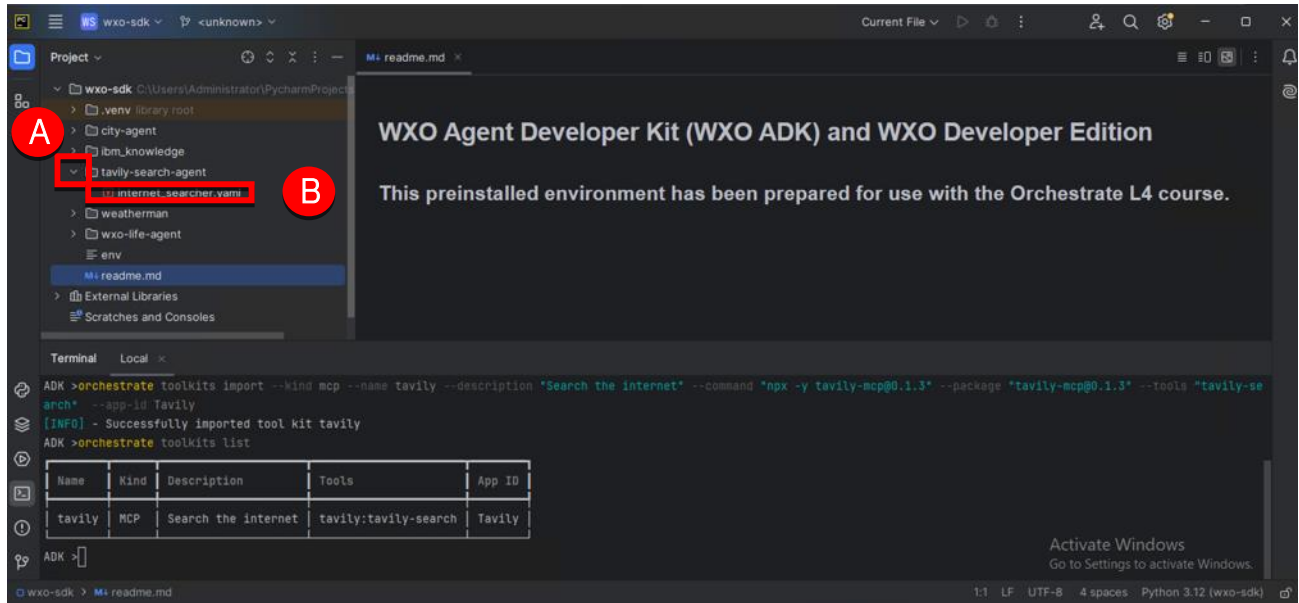


The toolkit is ready, and it contains the **tavily:tavily-search** tool as expected. You can now create the agent that will use this search tool.

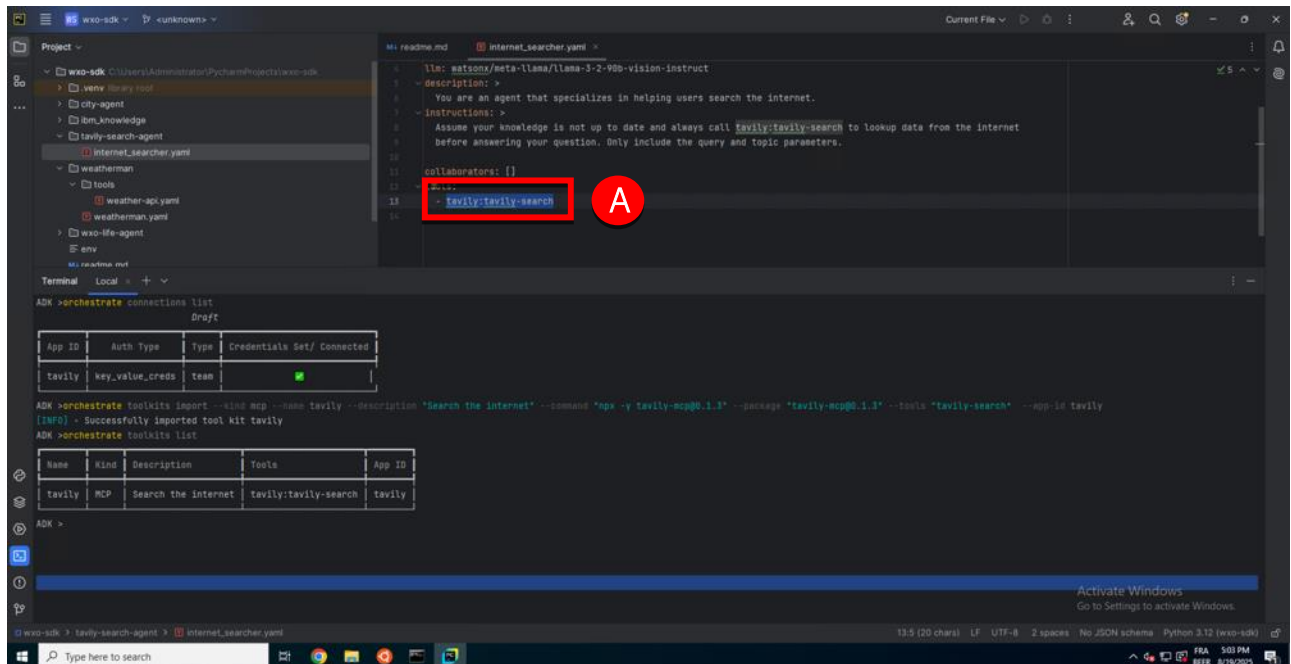
Create the Tavily agent

The Tavily_search agent is described in the internet_searcher.yaml file contained in your workspace.

1. Expand the **Tavily-search-agent** folder (A) then double click the **internet_searcher.yaml** file (B).

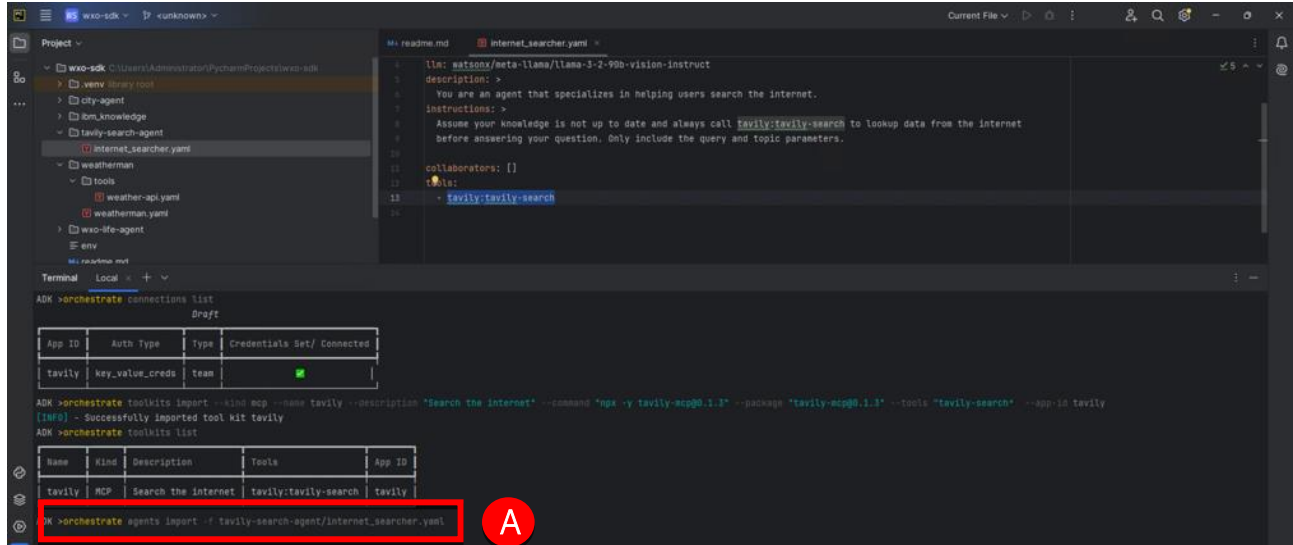


2. Review the agent and note that it will use the Tavily-search tool you have imported from Tavily (A).



3. Type the following command in the terminal to import the agent (A).

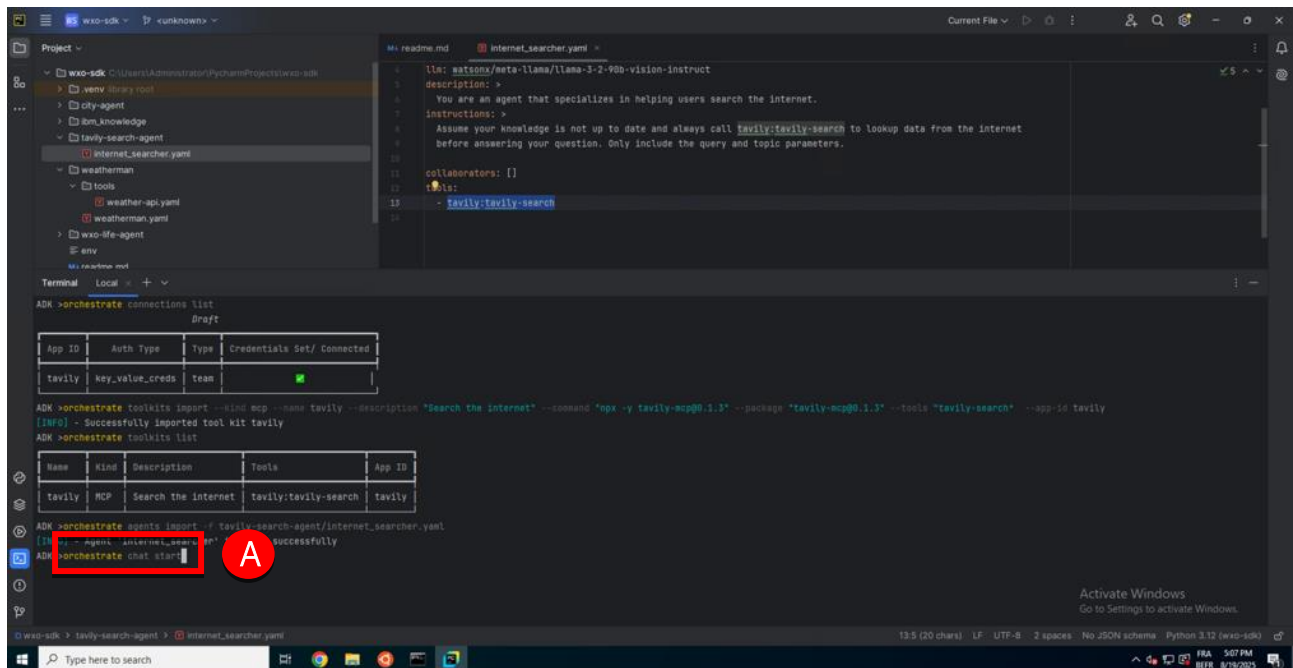
```
orchestrate agents import -f tavily-search-agent/internet_searcher.yaml
```



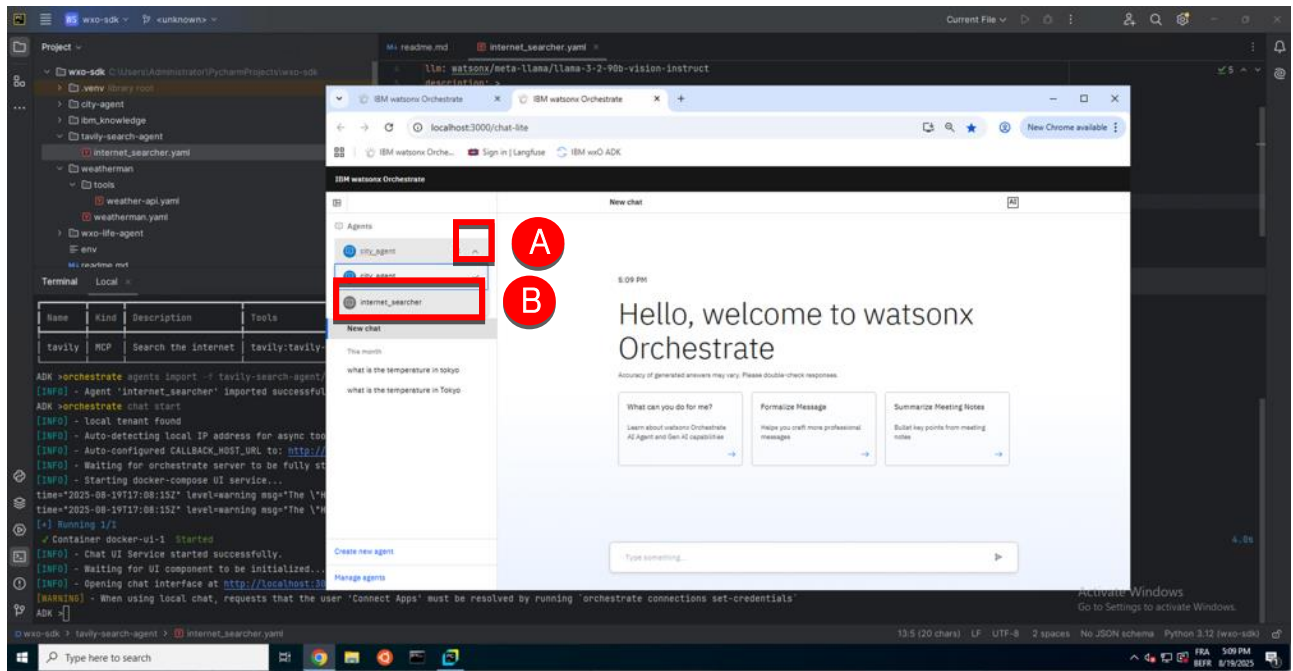
Preview the agent

1. Type the following command to start the chat interface (A).

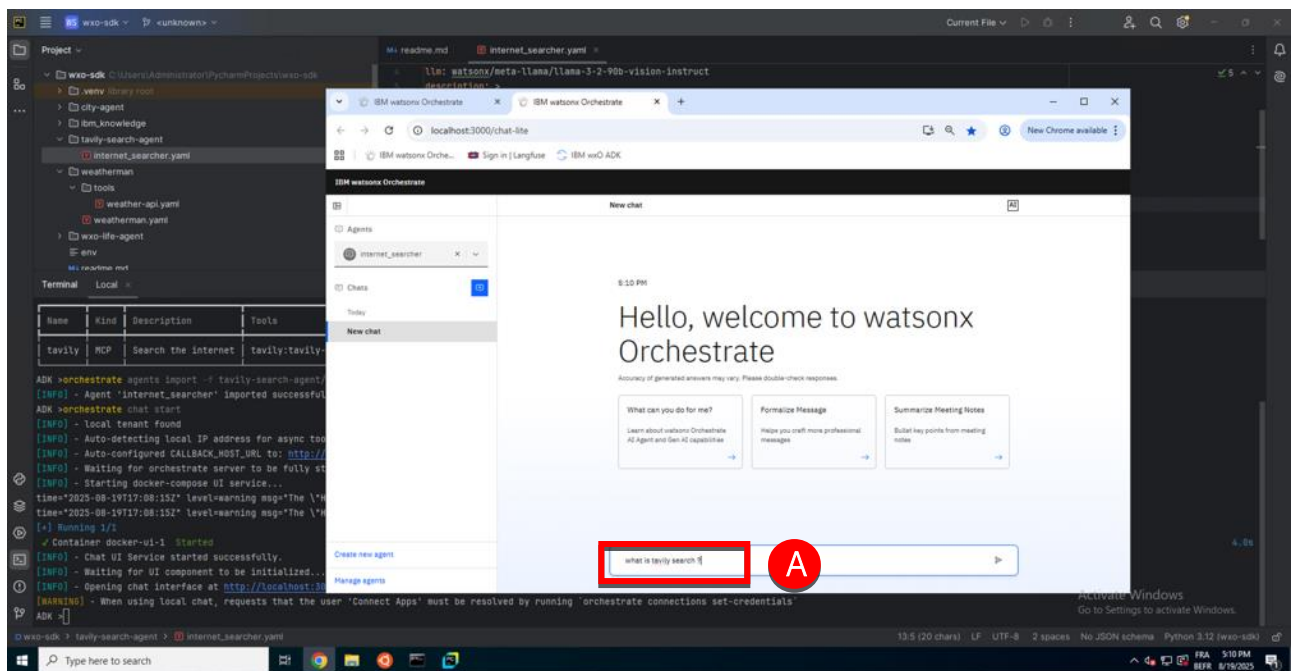
```
Orchestrate chat start
```



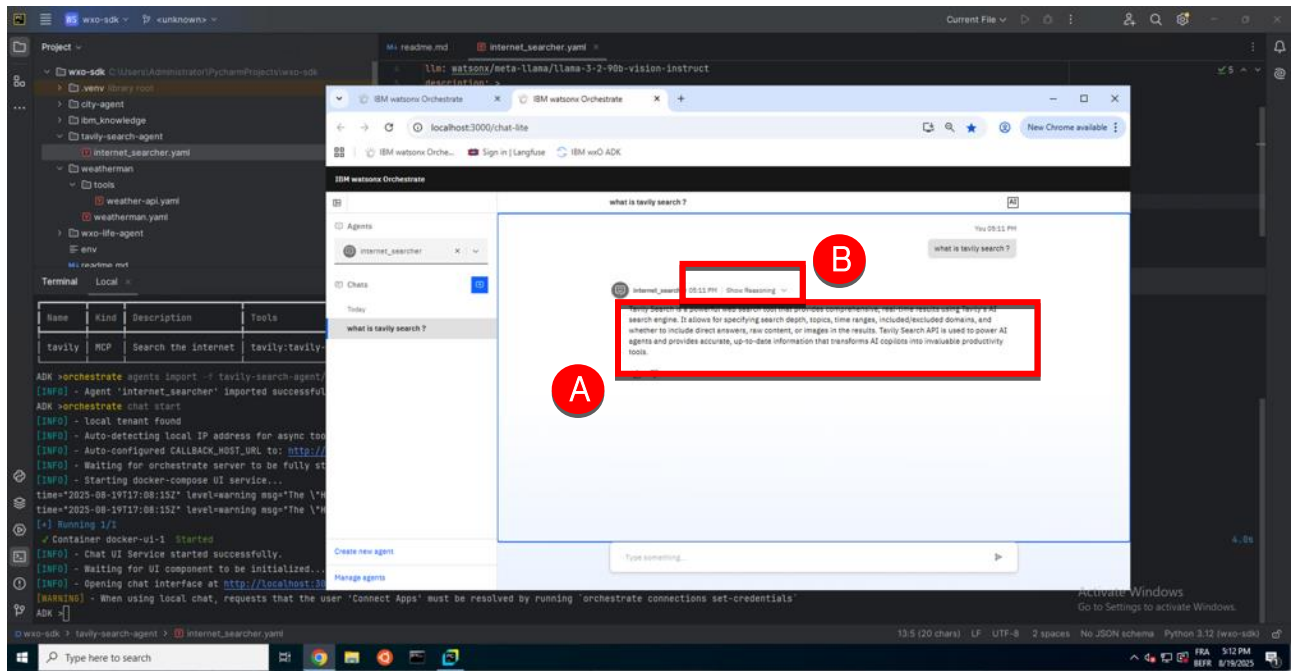
- Expand the agent selector (A) and select the **Internet_searcher** agent (B) from the list.



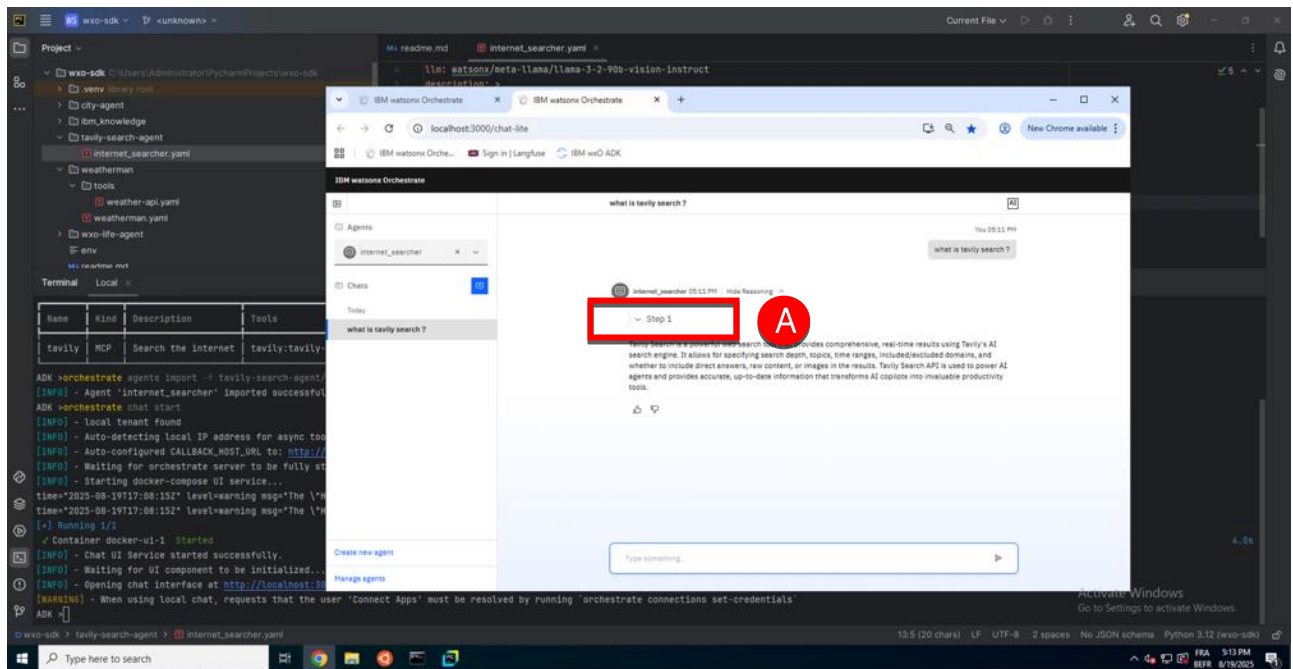
- Type 'what is Tavily search?' in the chat bar (A).



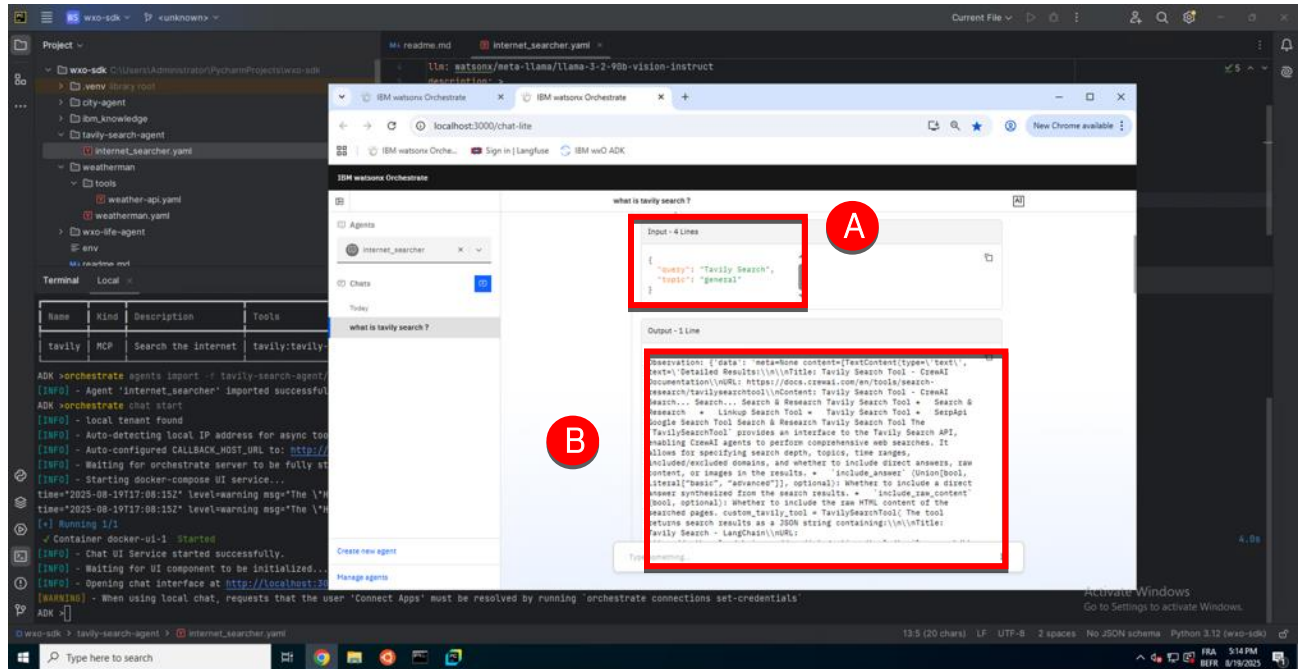
4. Note the response (A) then expand the **Show reasoning** menu (B).



5. Expand **Step 1** (A)



6. You can now review in detail how the Tavily search engine was invoked (A), how the response was sent back by the server and how the final answer was generated by the LLM (B).



Next step

Congratulations, you have reached the end of this lab. In a later lab you will learn how to use the ADK and SaaS environment together and connect your ADK CLI to the SaaS environment so that CLI commands are run against the SaaS environment.

As a next step you can:

- Play with the parameters of the Tavily search tool to see the impact of the query result and study how the agent will manage these parameters
- Import new tools from other MCP servers that you can find here:
 - <https://github.com/appcypher/awesome-mcp-servers>
 - <https://github.com/modelcontextprotocol/servers>

Summary

Add a section at the end to summarize what the user accomplished from the lab.

In this lab you have:

- Created an admin account
- Installed the product
- Tested the access
- Verified the setup

Appendix: Troubleshooting

Unable to create a connection

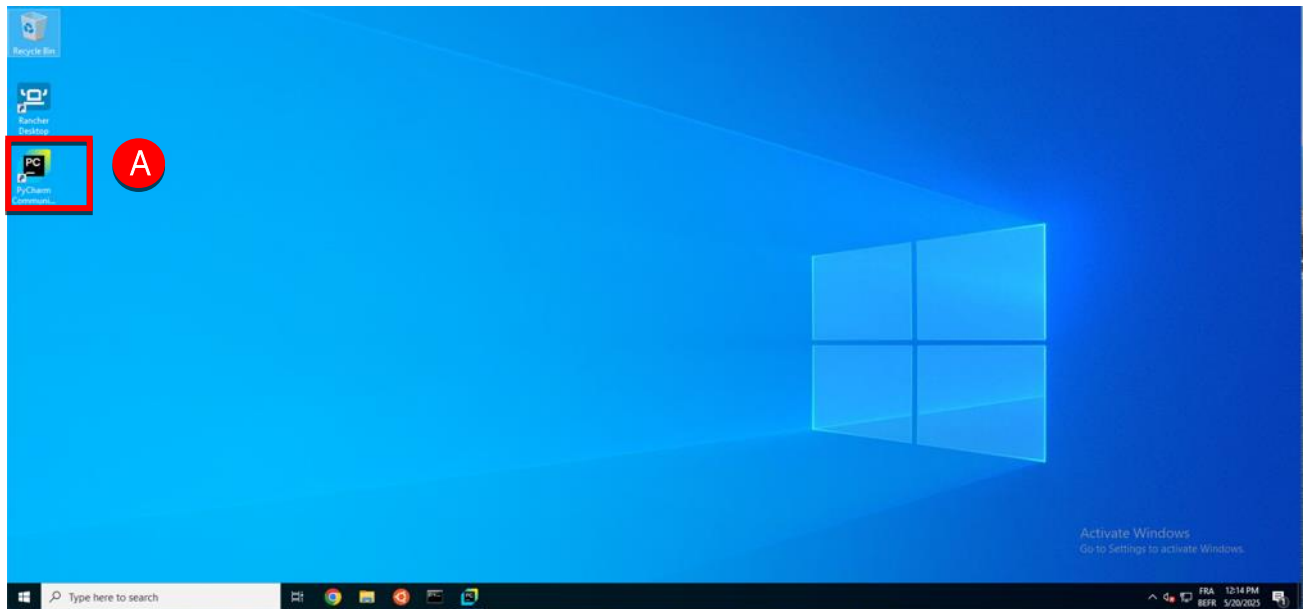
If you are unable to create a connection using watsonx Orchestrate running on IBM Cloud, try using a private browser window.

If the problem persists your SaaS environment may still be provisioning, if after an hour you are still unable to create a connection, delete the reservation and reserve another environment.

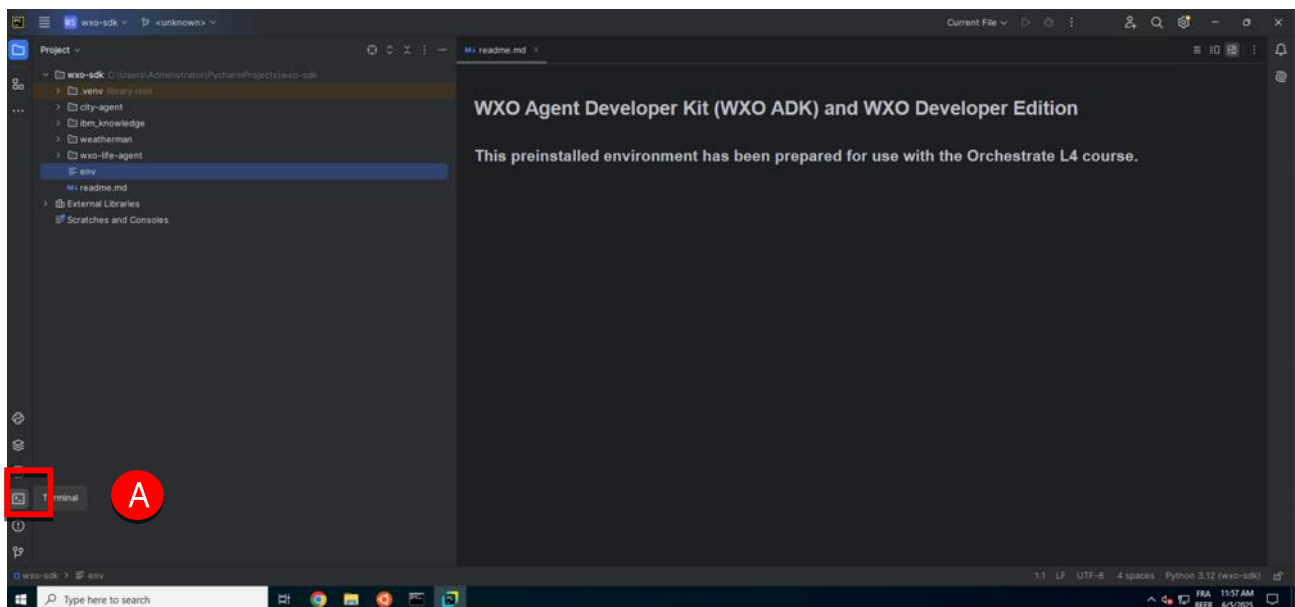
Appendix: Reset your environment

These steps will remove any existing tools and agents from your environment and restart the local watsonx Orchestrate server in readiness to start the lab.

1. Login your VM image containing your ADK (Please refer to the preparation guide for the detailed instructions).
2. Double-click the **PyCharm** icon (A), to open the editor. (PyCharm may be open already if you have used this environment to complete previous labs.)

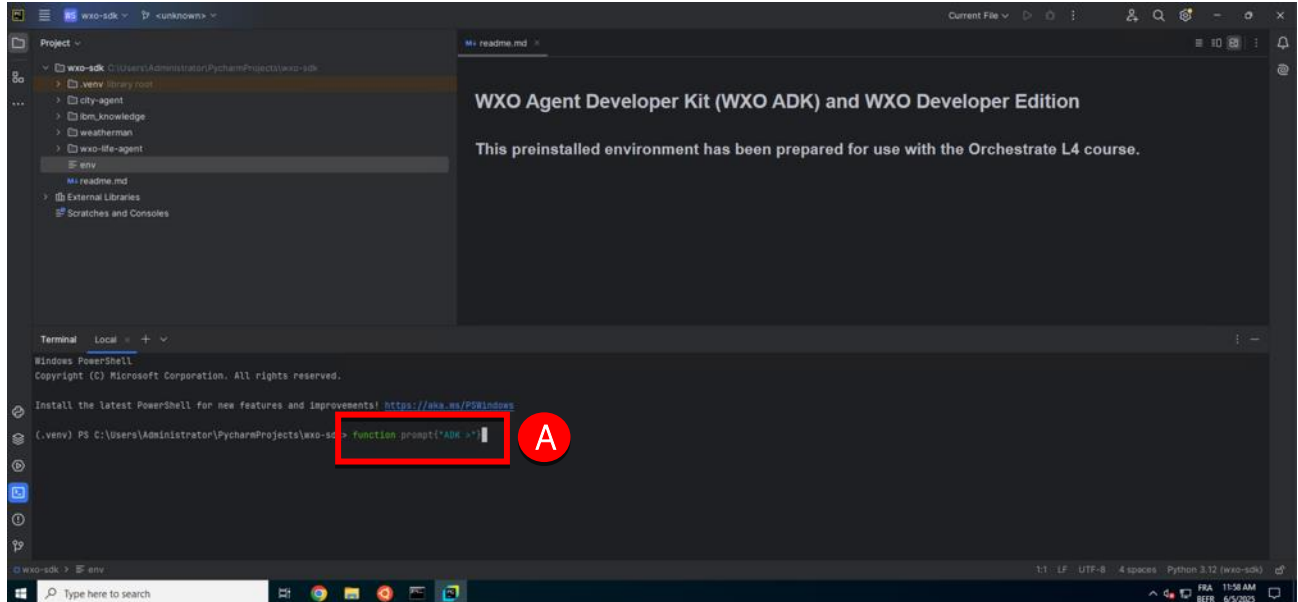


3. Click the terminal icon (A), to open the terminal (if required).



4. Enter the following command into the terminal (A), then press Enter:

```
function prompt{"ADK >"}
```



This changes the prompt and will improve the readability of commands used in this lab.

5. Enter the following command into the terminal to stop your chat service, then press Enter.

```
orchestrate chat stop
```

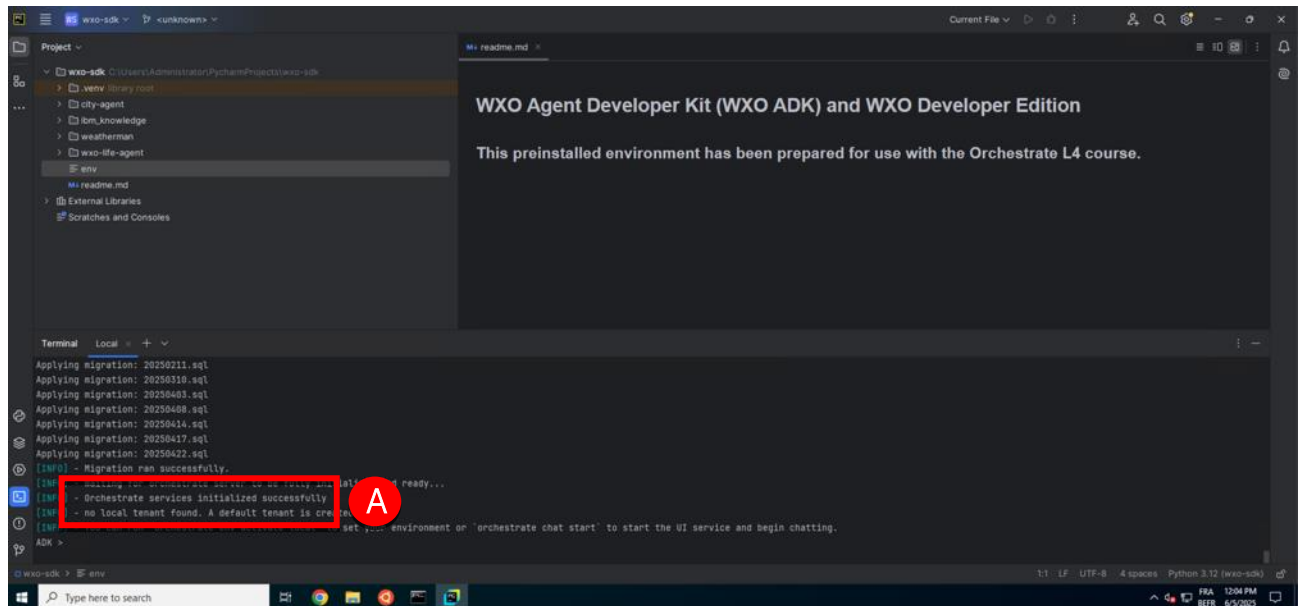
6. Enter the following command into the terminal to reset your server, then press Enter.

```
orchestrate server reset
```

7. Enter the following command into the terminal to restart your server, then press Enter.

```
orchestrate server start -l -e env
```

- When restart is complete (A) the terminal output will resemble the example below, restart can take 2-3 minutes to complete.



Note: If you encounter an error (A), please repeat the start command as some images require a little extra time to start:

